Three Architectures, One Routing Pattern: How VLA Fine-Tuning Reshapes Instruction Flow

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Abstract: Vision-language-action models (VLAs) convert pretrained vision-language backbones into robot controllers through fine-tuning, yet whether the resulting representational changes reflect general training dynamics or architecture-specific artifacts remains unknown. We conduct a systematic cross-architecture analysis of three VLAs built on distinct backbones (Llama-2-7B, Phi-3-mini, Qwen2.5-VL-3B) across four LIBERO task suites, comparing trained and untrained checkpoints with cross-instruction probes, attention profiling, R^2 action probes, and counterfactual instruction perturbation. We identify two training effects that generalize across architectures. First, R^2 exit collapse, where action-relevant information is selectively discarded at exit layers, consistently emerges when vision-language models are first fine-tuned into VLAs (6/6 cells across two architectures) and re-emerges for spatially demanding tasks when already-pretrained VLAs undergo further specialization (2/4 cells), suggesting that exit-layer reorganization scales with the representational distance between pretraining and target tasks; a chat-finetuned control using the same backbone shows no exit collapse ($\Delta R^2 \approx 0$), confirming this reorganization is specific to action-oriented training. Second, all trained VLAs exhibit substantial counterfactual sensitivity to instruction perturbation (Cohen's $d \geq 0.45$ across two architectures and eight suite combinations), though metric incomparability between trained (action MSE) and untrained (cosine distance) conditions precludes quantifying the trained–untrained gap as a single amplification factor. In contrast, early-layer instruction-dominant attention patterns trace to base-model priors rather than VLA training; one backbone exhibits saturated instruction attention even before fine-tuning. Together, these findings distinguish training-induced routing effects (exit collapse, counterfactual sensitivity) from inherited base-model priors (early-layer instruction attention), and their consistency across two VLM-to-VLA transitions with dramatically different starting points is compatible with VLA fine-tuning discovering shared functional organization constrained by the causal-decoder design.

Keywords: Vision-Language-Action models, representational analysis, attention routing, fine-tuning

1 Introduction

Vision-language-action models (VLAs) fine-tune pretrained language models on robot demonstrations to produce actions from visual observations and language instructions [1, 2]. In these decoder-only architectures, image tokens precede instruction tokens in the input sequence, and the causal attention mask prevents image tokens from attending to any token that follows them [3, 4]. Instruction information can reach action-generating positions only through later attention layers, where the readout token (the final position before action decoding) integrates context from all preceding tokens. The readout is thus the sole multimodal integration point by construction. How fine-tuning

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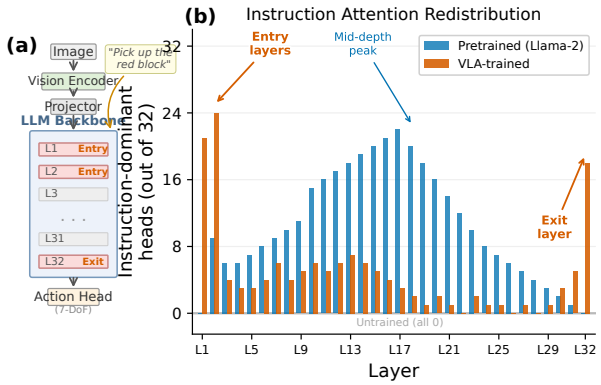


Figure 1: **VLA fine-tuning reorganizes action-relevant representations in a pattern consistent across two VLM-to-VLA transitions but absent before action training.** Three architectures are compared in trained and untrained conditions. Training-induced effects (R^2 exit collapse, counterfactual sensitivity) emerge at the VLM-to-VLA transition, while L1 instruction attention reflects inherited base-model priors.

creates or reshapes the attention patterns at this position, and whether the resulting routing reflects training dynamics or inherited backbone structure, remains unknown.

Behavioral studies provide indirect evidence that this question matters. VLAs exhibit visual dominance, relying disproportionately on image input [5], and language instructions appear largely unused by the action pathway [6]. These observations establish that VLAs underweight instructions but do not explain *how* instruction information reaches action-generating positions, *what* routing structure training creates versus what it inherits from the pretrained backbone, or *whether* observed patterns hold across architectures. All existing representational analyses of VLAs target a single architecture. No prior work tests whether instruction routing mechanisms are consistent across backbone families, and none decomposes training-induced effects from inherited base-model priors. Absent cross-architecture evidence, it is impossible to distinguish training dynamics that reflect the causal-decoder structure from architecture-specific artifacts of a particular language model.

We address this gap with a cross-architecture analysis of three VLAs: OpenVLA [2, 7] (Llama-2-7B, 32 layers), TracVLA [8] (Phi-3-mini, 32 layers), and NORA [9] (Qwen2.5-VL-3B, 36 layers), evaluated across four LIBERO [10] task suites in a 3×4 matrix (10/12 cells filled; §4.1). A critical asymmetry structures the comparison: OpenVLA's untrained checkpoint is the OXE-pretrained [11] VLA (970K episodes), while TracVLA and NORA baselines are pure VLMs. The six TracVLA/NORA cells test VLM-to-VLA transitions; the four OpenVLA cells test VLA-to-VLA transitions. We apply R^2 action probes, L1 attention profiling, and counterfactual instruction perturbation to each cell, with additional single-architecture analyses on OpenVLA (§3).

Three findings emerge (Figure 1). *First*, R^2 exit collapse—the drop from peak to exit layer—consistently marks VLM-to-VLA transitions (6/6 cells, sign test $p = 0.016$; TracVLA $\Delta R^2 = 0.14$–0.29; NORA up to 0.69). Under VLA-to-VLA conditions, collapse re-emerges for spatially demanding tasks (2/4 OpenVLA cells), suggesting it scales with pretraining-to-target representational distance. A chat-finetuned control on the same backbone shows $\Delta R^2 \approx 0$, confirming VLA-specificity (§4.3). *Second*, all trained VLAs exhibit counterfactual sensitivity ($d \geq 0.45$ across eight tested cells; empty perturbation ranks first in 7/8); metric incomparability between trained and untrained conditions precludes cross-metric ratios (§4.4). *Third*, L1 instruction attention reflects base-model priors: Phi-3-Vision saturates at 32/32 heads before training; Qwen2.5-VL allocates 13–14/16—both unchanged or reduced by VLA training (§4.5).

Our contributions are:

1. An empirical cross-architecture representational analysis framework for VLAs, combining factorial trained/untrained controls with token-level probing across a 3×4 architecture-suite matrix.

2. Identification of R^2 exit collapse as a VLA-specific marker of the VLM-to-VLA transition, consistent across two VLM-to-VLA architectures (TracVLA, NORA) with task-dependent confirmation in OpenVLA (2/4 suites), and absent in NLP finetuning of the same backbone.
3. A decomposition of VLA instruction routing into training-induced effects (exit collapse, counterfactual sensitivity) and inherited base-model priors (L1 instruction attention), grounded in cross-architecture trained/untrained contrasts.

2 Related Work

Vision-language-action models. VLAs fine-tune large language models on robot demonstrations to produce actions from visual and language input [1, 2, 7, 12, 8, 9, 13]. NLP instruction tuning is well-characterized behaviorally [14, 15], but the representational effects of VLA fine-tuning remain unexamined. Fine-tuning operates in a low intrinsic dimension [16] and can distort pretrained features [17]; existing analyses target single architectures and cannot separate training dynamics from backbone artifacts.

Mechanistic and behavioral analysis. Mechanistic interpretability has revealed residual stream communication [18], induction heads [19], and task circuits [20, 21]; causal tracing localizes factual recall [22]; attention probing shows heads encode syntactic relations [23] and most can be pruned [24]. In VLMs, recent work characterizes visual token redundancy [25] and progressive visual-textual alignment [26, 27]. Behavioral evaluations of VLAs reveal visual dominance [5, 28, 29]; mechanistic studies identify sparse semantic directions [6], interpretable features [30, 31], symbolic representations [32, 33], and reasoning erosion [34]. All prior representational analyses target a single architecture; our 3×4 cross-architecture matrix tests whether mechanisms replicate across backbone families.

Probing and similarity methodology. RSA [35], CKA [36], and linear probes [37, 38] provide complementary geometric and functional evaluation. We adopt RSA and MLP probes (CKA in appendix); the two metrics can diverge on the same data (Appendix A.4), and Cohen's d vs. cosine distance are not comparable across trained/untrained conditions (§4.4).

3 Method

We investigate how action-relevant representations are organized across three VLA architectures using complementary probing, attention, and perturbation analyses.

3.1 Architectures and Data

We study three VLAs that span distinct backbone families (Table 1). **OpenVLA** [2, 7] chains a frozen DinoSigLIP vision backbone [39, 40] (DINOv2 ViT-L/14 + SigLIP-SO400M), an MLP projector, and Llama-2-7B [41] (6.74B parameters, 32 layers, 32 multi-head attention [MHA] heads) for autoregressive 7-D action decoding. It is pretrained on Open X-Embodiment [11] (OXE; 970K episodes across multiple robot platforms). **TracVLA** [8] pairs a SigLIP encoder with Phi-3-mini [42] (3.8B parameters, 32 layers, 32 MHA heads). Spatial trace conditioning renders visual trajectory overlays on input images to provide spatial guidance for action prediction. TracVLA is trained on an OXE subset. **NORA** [9] uses Qwen2.5-VL-3B ($\sim$3B parameters, 36 layers, grouped-query attention [GQA] with 16 query heads and 2 KV heads) with the native Qwen2.5-VL vision encoder, which processes images at dynamic resolution without a separate projection stage.

Each architecture is evaluated in trained and untrained conditions. A critical asymmetry distinguishes the untrained baselines: OpenVLA's untrained checkpoint is the OXE-pretrained VLA, already fine-tuned for action prediction on 970K episodes, while TracVLA and NORA untrained checkpoints are pure vision-language models (Phi-3-Vision and Qwen2.5-VL) with no action training. OpenVLA trained-vs-untrained comparisons therefore measure a VLA-to-VLA transition;

Table 1: Architectures studied. The untrained baselines differ in kind: OpenVLA's baseline is an OXE-pretrained VLA that produces actions; TracVLA and NORA baselines are pure VLMs with no action training.

Arch.	Backbone	Layers	Attention	Vision Enc.	Untrained Baseline
OpenVLA	Llama-2-7B	32	32 MHA	DinoSigLIP → MLP proj.	OXE-pret. VLA
TracVLA	Phi-3-mini	32	32 MHA	SigLIP + spatial trace	Phi-3-Vision (VLM)
NORA	Qwen2.5-VL-3B	36	16q/2kv GQA	Qwen2.5-VL native	Qwen2.5-VL (VLM)

TracVLA and NORA comparisons measure VLM-to-VLA transitions. This asymmetry affects the interpretation of all trained/untrained comparisons throughout the paper. For OpenVLA, two additional conditions support deep-dive analyses (§4.2): the Llama-2 base language model (no VLA fine-tuning) and a random-weight initialization via `from_config()`.

We evaluate on LIBERO [10], comprising four task suites (Goal, Spatial, Object, Long-Horizon) with 10 tasks and 50 demonstrations each. The three architectures and four suites form a 3×4 matrix; 10 of 12 cells are filled (NORA Object and Long-Horizon lack available checkpoints).

3.2 Representation Extraction

From each architecture, we extract hidden states at every transformer layer from two token positions: (a) **image_mean**, the mean activation over image tokens, and (b) **readout**, the hidden state at the final pre-action token, which follows both image and instruction tokens in the causal sequence and can attend to all preceding context. L0 denotes the embedding layer output, L1 through LN the hidden states after each transformer layer ($N = 32$ for OpenVLA and TracVLA, $N = 36$ for NORA), with the final layer including the output layer norm. In all three architectures, image tokens precede instruction tokens; the causal mask prevents image tokens from attending to instructions by construction.

We cross two factors in a 2×2 factorial design: *training status* (trained vs. untrained) and *instruction presence* (with vs. without task description), applied to all 10 available cells in the cross-architecture matrix. OpenVLA additionally provides representations from the DinoSigLIP projector output (4,096-D), DINOv2 and CLIP external baselines, and the branch-split before fusion; these sources support the OpenVLA deep-dive analyses (§4.2).

3.3 Cross-Architecture Analyses

Three analyses are applied to every available cell in the 3×4 matrix.

R^2 action probes. Two-layer MLP probes [37] ($d_{\text{in}} \to 256 \to 7$, ReLU) are trained to predict 7-D action vectors from frozen representations at each layer (Adam, lr $= 10^{-3}$, batch size 256, 50 epochs, early stopping patience 5). Train/test splits follow demonstration boundaries (demos 0–39 vs. 40–49) to prevent within-trajectory leakage. We report per-task macro-averaged R^2 with bootstrap paired comparisons (10,000 resamples).

L1 attention head profiling. At L1, each attention head is classified as instruction-dominant ($>$ 50% of readout-token attention allocated to instruction tokens) or image-dominant. Results are reported as k/N instruction-dominant heads, where N is the total L1 head count: 32 for OpenVLA and TracVLA (MHA), 16 for NORA (GQA query heads). Sensitivity analysis at thresholds from 30% to 70% confirms ordering robustness (Appendix A.21.8).

Counterfactual instruction perturbation. The instruction is replaced with four conditions: empty string, wrong task from the same suite, shuffled tokens, and paraphrased instruction. A metric constraint arises from the trained/untrained asymmetry. Trained models produce action outputs; their sensitivity is measured via Cohen's d on action MSE (OpenVLA, TracVLA) or cosine distance on action representations (NORA). Untrained VLMs (Phi-3-Vision, Qwen2.5-VL) do not generate actions; their sensitivity is measured via cosine distance on hidden representations, a metric

with different scale and distributional properties. Cross-metric ratios between Cohen's d and cosine distance are therefore invalid (§4.4). OpenVLA's OXE-pretrained untrained checkpoint is the exception: because it produces actions, same-metric (Cohen's d) trained-vs-untrained comparison is possible for that condition.

OpenVLA additionally supports cross-instruction probes, representational similarity analysis (RSA) [35], and external baselines (DINOv2 [40], CLIP [43]); these single-architecture methods are detailed in Appendix A.1.

4 Experiments

4.1 Experimental Setup

We analyze three VLA architectures across four LIBERO [10] task suites (Goal, Spatial, Object, Long-Horizon) in a 3×4 cross-architecture matrix (Table 2). The architectures span distinct backbone families: OpenVLA [2, 7] (Llama-2-7B [41], 32 layers, 32 MHA heads), TracVLA [8] (Phi-3-mini [42], 32 layers, 32 heads), and NORA [9] (Qwen2.5-VL-3B, 36 layers, GQA with 16 query heads and 2 KV heads). Of the 12 matrix cells, 10 are filled; NORA Object and Long-Horizon lack available checkpoints. Each architecture is compared in trained and untrained conditions, with a critical asymmetry between untrained baselines: OpenVLA's untrained checkpoint is the OXE-pretrained [11] VLA (already fine-tuned for action prediction on 970K episodes), while TracVLA and NORA untrained checkpoints are pure vision-language models (Phi-3-Vision and Qwen2.5-VL) with no action training. Three analyses are applied to each cell (§3): R^2 action probes at all layers, L1 attention head profiling for instruction dominance, and counterfactual instruction perturbation. For OpenVLA LIBERO-Goal, $N = 63{,}228$ observations; other cells are at comparable scale. For trained models that produce action outputs, counterfactual sensitivity is measured via Cohen's d on action MSE. For untrained VLMs (Phi-3-Vision, Qwen2.5-VL) that do not produce actions, sensitivity is measured via cosine distance on hidden representations; these metrics operate on different scales and are not directly comparable (§4.4). OpenVLA's OXE-pretrained checkpoint produces actions, permitting same-metric (Cohen's d) comparison in that condition. Bonferroni correction details appear in Appendix A.29.

4.2 OpenVLA: Motivating Single-Architecture Analysis

Before testing cross-architectural generality, we establish three core phenomena on OpenVLA [2] (full analysis in Appendix A.2). At L1, three conditions form a gradient: random-weight initialization allocates 0/32 instruction-dominant heads, Llama-2 base [41] allocates 9/32, and VLA-trained allocates 24/32—confirming that VLA fine-tuning creates instruction routing beyond language-model priors. Counterfactual perturbation produces graded action divergence ordered by semantic distance ($d = 0.71$–0.98 across Goal and Spatial suites; empty $>$ wrong $>$ shuffled $>$ paraphrased), and dose-response ablation of instruction-dominant heads yields monotonic trajectory divergence ($d = 0.48$–0.64; Appendix A.11). R^2 probing localizes training effects at the readout token ($+0.069$) rather than image_mean ($+0.015$), and training stabilizes readout instruction classification at 100% through L31 (untrained: 82% decay). These three phenomena—L1 instruction attention, counterfactual sensitivity, and readout-localized training effects—motivate the cross-architecture tests that follow.

4.3 Cross-Architecture R^2 Exit Collapse

We define R^2 exit collapse as $\Delta R^2 = R^2_{\text{peak}} - R^2_{\text{exit}}$, the drop in action-probe accuracy from peak layer to exit layer (Table 2). All six VLM-to-VLA cells show clear collapse: TracVLA [8] trained ΔR^2 ranges from 0.138 to 0.294 (untrained: 0.009–0.048), and NORA [9] reaches $\Delta R^2 = 0.691$ on Goal, where exit-layer R^2 drops below zero (Figure 2). Under VLA-to-VLA conditions, OpenVLA [2] shows task-dependent collapse: Spatial ($\Delta R^2 = 0.110$ trained vs. 0.052 untrained) and Object (0.121 vs. 0.076) exhibit clear collapse, while Goal (0.016 vs. 0.033) and Long (0.004 vs.

Table 2: Cross-architecture summary of R^2 exit collapse (ΔR^2: peak-to-exit drop in action-probe R^2) and counterfactual (CF) sensitivity (effect size of top perturbation condition). All 10 filled cells in the 3×4 architecture $\times$ suite matrix are shown. Top CF condition is "empty" for all trained cells except OpenVLA Long ("shuffled"). **Trained/untrained asymmetry**: OpenVLA untrained is an OXE-pretrained VLA; TracVLA and NORA untrained are pure VLMs with no action training. **Metric note**: trained CF uses Cohen's d (OpenVLA, TracVLA) or cosine distance (NORA); untrained CF uses cosine distance except where noted. Cross-metric ratios are not meaningful (§4.4).

		ΔR^2		CF (top cond.)	
Arch.	Suite	Trained	Untr.	Trained	Untr.
OpenVLA	Goal	0.016	0.033	0.98	—
OpenVLA	Spatial	0.110	0.052	1.17	0.94‡
OpenVLA	Object	0.121	0.076	1.21	0.57*
OpenVLA	Long	0.004	0.016	0.56†	1.05‡
TracVLA	Goal	0.228	0.048	1.52	0.15*
TracVLA	Spatial	0.138	0.016	1.33	0.18*
TracVLA	Object	0.294	0.009	1.21	0.19*
TracVLA	Long	0.198	0.028	1.47	0.17*
NORA	Goal	0.691	0.044	sat.§	0.08*
NORA	Spatial	0.302	0.037	0.28*	—
NORA	Obj./Long	*no checkpoints available*			

Trained CF: Cohen's d (OpenVLA, TracVLA) or cosine distance* (NORA).
*Cosine distance on hidden representations (different scale from d).
†Top cond. is "shuffled." ‡Cohen's d; untrained is an OXE-pretrained VLA.
§Cosine distances saturated at 1.0; earlier layers differentiate.

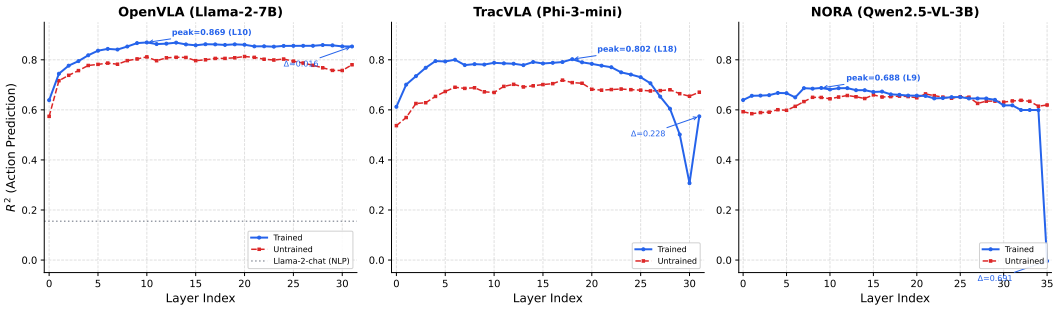


Figure 2: Per-layer R^2 action-probe profiles for three architectures, each comparing trained (solid) and untrained (dashed) checkpoints. All six VLM-to-VLA cells (TracVLA and NORA, 6/6) exhibit pronounced peak-to-exit collapse ($\Delta R^2 = 0.14$–0.69), while their untrained VLM baselines show flat profiles ($\Delta R^2 \leq 0.05$). For OXE-to-LIBERO cells, collapse is task-dependent: Spatial ($\Delta R^2 = 0.11$) and Object (0.12) show collapse, while Goal and Long do not; OpenVLA's untrained baseline is an OXE-pretrained VLA that has already undergone exit-layer reorganization. One-sided sign test confirms the VLM-to-VLA direction across all six cells ($p = 0.016$).

0.016) do not—consistent with collapse scaling with the representational distance between pretraining and target task. A sign test across all six VLM-to-VLA cells confirms the direction ($p = 0.016$); across all ten cells, eight show trained $>$ untrained ($p = 0.055$). A chat-finetuned control (Llama-2-7b-chat, same backbone as OpenVLA) shows $\Delta R^2 = 0.000$, confirming that exit collapse is specific to action-oriented training rather than a generic fine-tuning artifact.

4.4 Cross-Architecture Counterfactual Sensitivity

All trained VLAs exhibit substantial counterfactual sensitivity to instruction perturbation. Across eight cells where Cohen's d is applicable (OpenVLA and TracVLA, four suites each), every perturbation condition produces $d \geq 0.45$ (Table 3 in Appendix A.3). The "empty" condition ranks first in 7 of 8 cells ($d = 0.98$–1.52); "paraphrased" ranks last in 7 of 8, indicating that VLAs encode instruction content beyond token-level matching. TracVLA produces systematically higher effect

sizes ($d = 0.97$–1.52) than OpenVLA (0.45–1.21). NORA uses cosine distance (saturated at 1.0 on Goal; peak 0.275 on Spatial). Metric incomparability between trained (Cohen's d) and untrained (cosine distance) conditions precludes cross-metric amplification ratios (see Appendix A.3 for metric discussion).

Two OpenVLA cells permit same-metric trained/untrained comparison, since the OXE-pretrained baseline also produces actions. On Spatial, training increases sensitivity across all four conditions ($+0.05$ to $+0.24$), aligning with R^2 exit collapse ($\Delta R^2 = 0.110$). On Long-Horizon, the pattern inverts: the OXE-pretrained model ($d = 1.05$) exceeds the LIBERO-trained model ($d = 0.56$), consistent with near-zero collapse ($\Delta R^2 = 0.004$) and the general-purpose model maintaining broader instruction responsiveness.

4.5 L1 Attention Reflects Base-Model Priors

L1 instruction-dominant head counts across architectures reveal a pattern that contrasts with the training-induced effects documented in §4.3 and §4.4. TracVLA untrained (Phi-3-Vision) allocates 32/32 heads to instruction-dominant attention across all four suites, saturating the capacity of its 32-head L1 before any VLA training. Training slightly *reduces* this count to 27–30/32, with freed heads presumably repurposed for action-related computation. NORA untrained (Qwen2.5-VL) allocates 13–14 of its 16 L1 heads to instruction dominance (81–88%), and training leaves this essentially unchanged at 12–13/16. OpenVLA shows 24–26/32 instruction-dominant heads in the OXE-pretrained baseline; trained values are similar (14–24/32), with the largest change in Long-Horizon where training reduces the count from 25 to 14 heads (Figure 3).

If VLA training drove L1 instruction routing, untrained VLMs should show low attention that increases after training—as observed in OpenVLA's random-weight condition (0/32; §4.2). Instead, both untrained VLMs already saturate or near-saturate, and training leaves these counts unchanged or slightly reduced. L1 instruction attention is an inherited base-model prior, not a product of VLA fine-tuning.

4.6 Synthesis: Training-Induced vs. Inherited Routing

§4.3–4.5 distinguish two classes of routing effects: training-induced (R^2 exit collapse, counterfactual sensitivity) and inherited (L1 instruction attention). Training-induced effects emerge at the VLM-to-VLA transition (6/6 cells, $p = 0.016$) and partially re-emerge under VLA-to-VLA conditions (2/4 cells); their consistency across two architectures with dramatically different base-model priors is compatible with task-structure constraints of VLA fine-tuning rather than architecture-specific artifacts (§5).

5 Discussion

Synthesis. VLA fine-tuning produces two training-induced representational changes—R^2 exit collapse and counterfactual sensitivity—that concentrate at the readout token (training shifts readout R^2 by $+0.069$ vs. image_mean by $+0.015$). L1 instruction attention is inherited from base-model pretraining (Phi-3-Vision: 32/32, Qwen2.5-VL: 13–14/16 before training). The NLP control (Llama-2-7b-chat, $\Delta R^2 = 0.000$) confirms exit collapse is specific to action-oriented training. VLA fine-tuning does not build instruction routing from scratch; it reorganizes downstream representations while preserving early-layer attention structure.

Cross-architecture consistency and behavioral relevance. Two VLM-to-VLA transitions converge on the same training-induced effects (6/6 cells, $p = 0.016$), compatible with task-structure constraints of causal-decoder VLA design. OpenVLA confirms in a task-dependent manner under VLA-to-VLA conditions (2/4 cells). The cross-architecture matrix suggests a three-tier gradient (VLM $\rightarrow$ general VLA $\rightarrow$ task-specific VLA), but Tier 2 contains only one architecture ($n = 1$), limiting this to an observational synthesis (Appendix A.32). Correlating trained ΔR^2 with LIBERO

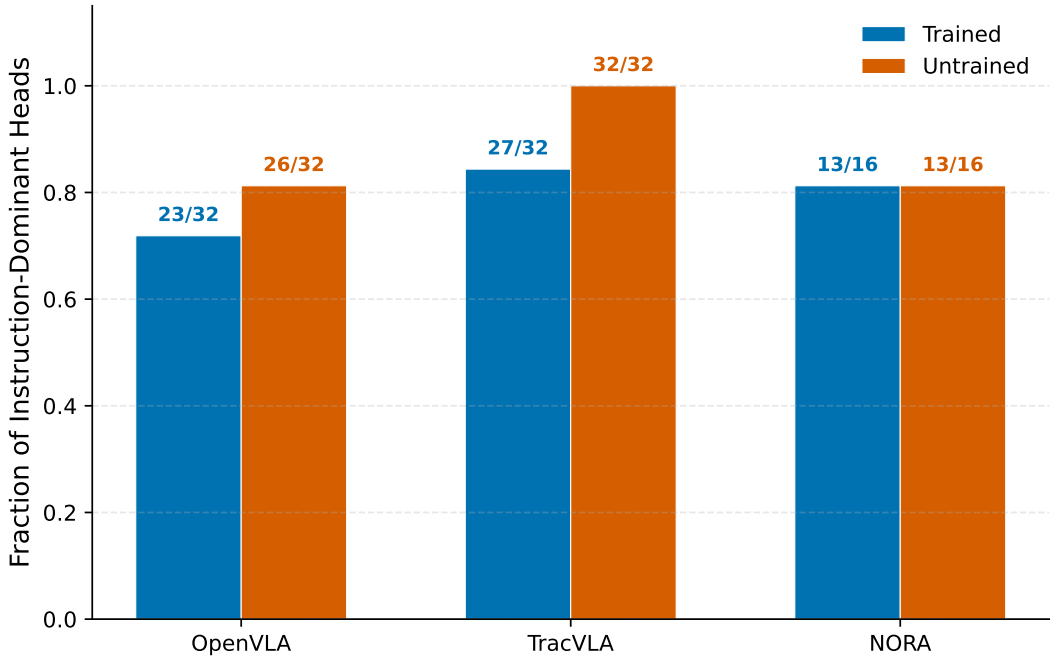


Figure 3: L1 instruction-dominant head counts across three architectures in trained and untrained conditions. L1 instruction attention is a base-model property: TracVLA untrained (Phi-3-Vision) saturates at 32/32 heads before any VLA training, and training *reduces* this to 27–30/32. NORA (Qwen2.5-VL, 16-head GQA) shows 13–14/16 in both conditions, unchanged by training. OpenVLA ranges from 14–24/32 trained vs. 24–26/32 untrained (OXE-pretrained). VLA fine-tuning does not create L1 instruction routing; it inherits and occasionally diminishes what the pretrained backbone provides.

success rates yields a positive but non-significant trend ($\rho = 0.583$, $p = 0.10$, $n = 9$); within-suite comparisons are uniformly positive but sample size limits statistical power (Appendix A.32).

Limitations. Cross-architecture evidence rests on two VLM-to-VLA transitions; OpenVLA tests a different transition type. All experiments use LIBERO; benchmarks with richer language grounding (CALVIN, Language-Table) are needed. We analyze only causal-decoder VLAs; flow-matching [12] and diffusion [44] architectures may differ. NORA lacks 2 of 4 suite checkpoints, and each architecture is analyzed at a single trained checkpoint. Extended limitations appear in Appendix A.32.

Conclusion. We analyzed three VLA architectures (OpenVLA, TracVLA, NORA) across four LIBERO task suites, comparing trained and untrained checkpoints to decompose VLA fine-tuning into training-induced and inherited representational effects. Two training-induced effects are consistent across two VLM-to-VLA transitions: R^2 exit collapse, where action-relevant information encoded at intermediate layers is selectively discarded at exit layers (6/6 cells, sign test $p = 0.016$), and counterfactual sensitivity, where all trained models respond to instruction perturbation (Cohen's $d \geq 0.45$ across eight tested cells). Exit collapse universally marks the VLM-to-VLA transition and re-emerges for spatially demanding tasks under VLA-to-VLA conditions (2/4 cells), suggesting that exit-layer reorganization scales with the representational distance between pretraining and target tasks; an NLP finetuning control on the same backbone ($\Delta R^2 = 0.000$) confirms this effect is specific to action-oriented training. In contrast, L1 instruction-dominant attention is inherited from base-model pretraining: one backbone (Phi-3-Vision) saturates at 32/32 heads before any action training. Metric incomparability between trained (action MSE) and untrained (cosine distance) conditions precludes quantifying counterfactual amplification factors, and the task-dependent nature of VLA-to-VLA collapse (2/4 OpenVLA suites) indicates that exit reorganization is modulated by pretraining-target distance rather than being universally triggered by further fine-tuning. Extending this framework to flow-matching and diffusion VLAs with cross-attention, to benchmarks with

richer language grounding, and to intermediate training checkpoints are priorities for future work. Code and data will be released upon acceptance.

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Use of Generative AI. Generative AI tools (Claude, GPT-4) assisted with code implementation and manuscript editing; all scientific content was produced by the authors.

A Supplementary Material

Unless noted otherwise, all analyses use OpenVLA-7B (`openvla/openvla-7b` on HuggingFace) evaluated on LIBERO-Goal (10 tasks $\times$ 50 demonstrations, $N = 63{,}228$ observations after excluding terminal frames). Model conditions, representation extraction, metric definitions, and probe architectures follow §3. Cross-benchmark experiments (§A.17) use LIBERO-Spatial with the same protocol (10 tasks $\times$ 50 demonstrations). All experiments ran on 2$\times$ NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (96 GB each); random seed is 42 unless stated otherwise.

A.1 OpenVLA-Specific Analysis Methods

Three additional analyses are applied to OpenVLA to characterize instruction encoding and representational geometry.

Cross-instruction probes. Each visual frame is paired with all 10 LIBERO-Goal task instructions, producing 10 representation vectors per frame per layer. A 10-way linear classifier identifies which instruction was present, applied separately at image_mean and readout, with leave-one-out cross-validation (LOOCV) over 10 folds.

Representational similarity analysis (RSA). RSA [35] quantifies geometric alignment between representations and the action space. For each task, we compute pairwise representational dissimilarity matrices (RDMs) using cosine distances for representations and Euclidean distances for 7-D action vectors, from within-task observation pairs across 50 demonstrations. Alignment is measured via Spearman ρ between upper-triangular RDM entries; we report mean ρ across 10 tasks with 95% confidence intervals from 10,000 bootstrap resamples at the task level [45]. We prefer RSA over CKA [36]: RSA captures relative distance structure via scale-invariant rank correlation, suited to comparing high- and low-dimensional spaces (4,096-D vs. 7-D), while CKA measures global subspace overlap that can remain high when geometric structure differs. Both metrics are reported in §A.4; their divergence (§4.6) confirms that neither alone fully characterizes representational structure.

External baselines. DINOv2 ViT-L/14 [40] (`cls_token`/`patch_mean`) and CLIP ViT-L/14 [43] (`patch_mean`) provide self-supervised and vision-language pretrained reference points for probing and RSA.

A.2 OpenVLA Single-Architecture Deep Dive

This section provides the full single-architecture analysis summarized in §4.2.

Cross-instruction probes. Cross-instruction probes validate the analysis framework on OpenVLA [2]: as predicted by the causal mask, image_mean instruction classification stays at chance (10%) across all 32 layers in all conditions (full profile in §A.21.1). Trained readout achieves 100% 10-way classification from L1 onward and maintains this through L31 (LOOCV-confirmed;

§A.21.2). Untrained readout also reaches 100% at L1 but decays to 82% by L31; the readout token is thus the primary multimodal integration site, and training stabilizes its instruction encoding across depth. All patterns reproduce with a linear probe (§A.13).

L1 attention gradient. At L1, three conditions form a gradient of instruction attention: untrained networks allocate 0/32 instruction-dominant heads (11.7% text attention, $1.5\times$ text-to-image ratio), Llama-2 base [41] allocates 9/32 (36.5%), and VLA-trained allocates 24/32 (57.7%, $38.2\times$ text-to-image ratio; §A.22). The 32-layer attention profile reveals qualitatively distinct organizations: Llama-2 base concentrates instruction attention at mid-depth (L9–L16: 15–22 heads), while VLA fine-tuning produces an entry-exit bookend pattern with L0–L1 at 21–24 heads, L31 at 18, and intermediate layers at 0–11 (§A.21.8). Head counts are computed at a 50% threshold; sensitivity analysis (30%–70%) confirms ordering robustness. A fourth condition, OpenVLA fine-tuned on LIBERO, preserves L1 entry concentration (22/32 heads) but loses L31 exit re-engagement entirely (0 vs. 18 heads for the OXE-pretrained model), consistent with exit-layer attention serving multi-task flexibility rather than task-specific routing. The redistribution beyond Llama-2 base priors confirms that VLA fine-tuning creates new instruction routing rather than merely inheriting language-model attention.

Counterfactual perturbation. Four perturbation types across 10 LIBERO-Goal tasks produce action divergence ordered by token novelty ($\rho = 0.8$): empty instruction ($d = 0.978$), wrong task (0.924), shuffled tokens (0.763), paraphrased (0.707). The shuffled-exceeds-paraphrased ordering, despite identical token sets, indicates that positional sensitivity dominates token-level novelty. The effect replicates across LIBERO-Spatial (10 tasks) with the same ordering: empty 0.942, wrong 0.851, shuffled 0.833, paraphrased 0.806 (Table 8). Across both suites (20 tasks), the gradient spans $d = 0.71$–0.98. Spatial instructions differ only in prepositions, yet wrong-task perturbation still produces $d = 0.851$; instruction-dominant heads route fine-grained spatial information, not merely task identity.

Dose-response ablation. Dose-response ablation of L1 instruction-dominant heads produces monotonic trajectory divergence: d rises from 0.482 to 0.639 as 4–32 heads are progressively ablated, with steep increase through 12 heads and plateau beyond 20 (§A.11). Ablating all 24 instruction-dominant heads alters 72.3% of action tokens (token-level $d = 0.65$, trajectory-level $d = 2.943$); control ablation of 8 random heads yields $d = 0.41$ (29.3% tokens affected), concentrated on the gripper dimension. The routing effect is task-dependent: instruction-discriminative tasks show $2.5\times$ larger $|\Delta R^2|$ under L1 ablation than visually-sufficient tasks ($p = 0.033$; §A.25).

R^2 localization. R^2 action probing localizes training effects: trained readout peaks at $R^2 = 0.678$ (L13), untrained readout peaks at 0.609 (L32), and the training effect concentrates at the readout ($+0.069$) rather than at image_mean ($+0.015$; §A.9). These effects survive Bonferroni correction across all 63 tests (§A.29).

A.3 Counterfactual Sensitivity Details

Table 3 reports per-condition counterfactual rankings for all eight cells where Cohen's d is available.

Metric constraints on trained–untrained comparison. Trained models produce action outputs, enabling Cohen's d on action MSE (OpenVLA, TracVLA) or cosine distance on action tokens (NORA). Untrained VLMs (Phi-3-Vision, Qwen2.5-VL) do not generate actions; their sensitivity is measured via cosine distance on hidden representations—a metric with different scale, units, and distributional properties. Ratios between Cohen's d and cosine distance are therefore methodologically invalid, and we do not report cross-metric amplification factors. Qualitatively, untrained VLMs show weak sensitivity: TracVLA untrained (Phi-3-Vision) produces peak cosine distances of 0.15–0.19; NORA untrained (Qwen2.5-VL) yields 0.081 on Goal.

Table 3: Counterfactual condition ranking for trained models (Cohen's d; higher = greater action divergence under instruction perturbation). Conditions: E = empty instruction, W = wrong task, S = shuffled tokens, P = paraphrased. "Empty" ranks first in 7 of 8 cells; OpenVLA Long is the exception ("shuffled" first). NORA cells use cosine distance and are reported in §4.4.

Arch.	Suite	1st	2nd	3rd	4th
OpenVLA	Goal	E 0.98	W 0.92	S 0.76	P 0.71
OpenVLA	Spatial	E 1.17	S 0.98	W 0.88	P 0.81
OpenVLA	Object	E 1.21	W 1.19	S 1.12	P 0.90
OpenVLA	Long	S 0.56	W 0.55	E 0.50	P 0.45
TracVLA	Goal	E 1.52	W 1.40	S 1.34	P 1.21
TracVLA	Spatial	E 1.33	S 1.29	P 1.18	W 1.17
TracVLA	Object	E 1.21	S 1.16	W 1.06	P 0.97
TracVLA	Long	E 1.47	W 1.42	S 1.39	P 1.16

NORA details. NORA Goal produces saturated exit-layer cosine distances of 1.0 for all conditions, with differentiation visible only at earlier layers (L24: wrong $= 0.123$, empty $= 0.098$). NORA Spatial produces a peak cosine distance of 0.275 for the "wrong" condition.

A.4 CKA Analysis Method

We employ Centered Kernel Alignment (CKA) [36] alongside RSA as a complementary perspective. While RSA captures local geometric alignment via rank-order preservation, CKA measures global representational similarity through kernel-based comparisons.

We compute both *linear CKA*, which uses a linear kernel and is sensitive to second-order statistics, and *RDM kernel CKA*, which operates on the same distance matrices as RSA but aggregates them differently. Using both variants tests whether conclusions about representation structure are robust to the choice of similarity metric.

Linear CKA between high-dimensional representations (4,096-D for LLM layers) and the 7-D action space is subject to a known dimension mismatch: the linear kernel in the low-dimensional space may not capture the same structure as in the high-dimensional space. RDM kernel CKA avoids this issue by operating on distance matrices of equal size regardless of the original feature dimensionality.

A.5 Metric Sensitivity: RSA versus CKA

Linear CKA [36] reverses the RSA ordering: untrained LLM (CKA $= 0.089$) $>$ trained LLM (0.083) $>$ DinoSigLIP+Projector (0.081) $>$ DINOv2 (0.066), partially inverting the self-supervised advantage (Figure 4). RDM kernel CKA partially resolves this discrepancy: trained (0.260) $\approx$ Projector (0.257) $>$ untrained (0.246) $>$ DINOv2 (0.241). This divergence reflects the known sensitivity of representation similarity metrics to different geometric properties.

At the per-layer level, linear CKA between image_mean and the action space rises monotonically from L0 (0.081) to L32 (0.098), with no inverted-U pattern. RSA captures rank-order geometric alignment; CKA captures global linear structure. The dissociation indicates that the geometric refinement in early layers operates on the relative arrangement of representations (affecting RSA) without altering the overall covariance structure (captured by CKA).

A.6 Branch-Split Detailed Analysis

To disentangle the contributions of the two encoder streams within DinoSigLIP, we extract representations from the DINOv2 branch (1,024-D) and SigLIP branch (1,152-D) separately before fusion. The two branches yield near-identical RSA: DINOv2 branch $\rho = 0.619$, SigLIP branch $\rho = 0.621$. Fusion produces a representation ($\rho = 0.634$) that exceeds either branch alone; concatenation captures complementary structure. The MLP projector then compresses this signal: DinoSigLIP+Projector achieves $\rho = 0.615$, a drop of $\Delta = 0.019$ from the fused 2,176-D input.

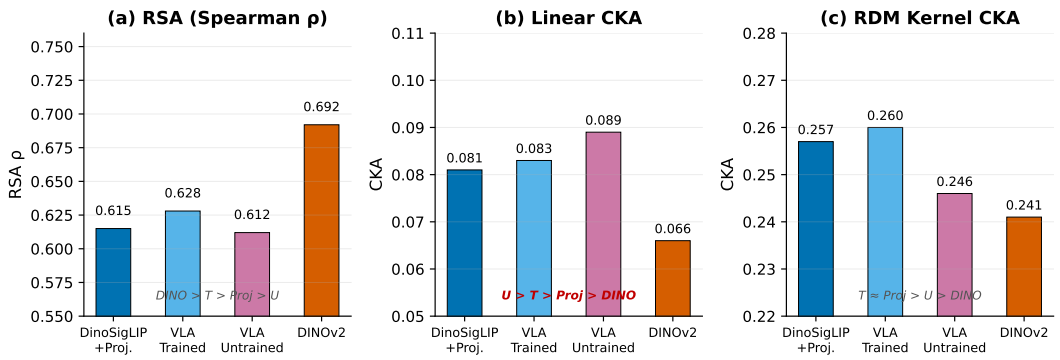


Figure 4: Three similarity metrics across representation sources. **(a)** RSA ranks DINOv2 cls (0.692) > Fused (0.634) > Projector (0.615). **(b)** Linear CKA reverses the ordering: Untrained (0.089) > Trained (0.083) > Projector (0.081) > DINOv2 (0.066). **(c)** RDM kernel CKA: Trained (0.260) ≈ Projector (0.257) > Untrained (0.246) > DINOv2 (0.241). The metric divergence underscores that RSA and CKA capture different aspects of representational structure.

A.7 RSA Across Representation Sources

Figure 5 shows RSA with the action space across all representation sources; the ordering matches the RSA results reported in §4.6.

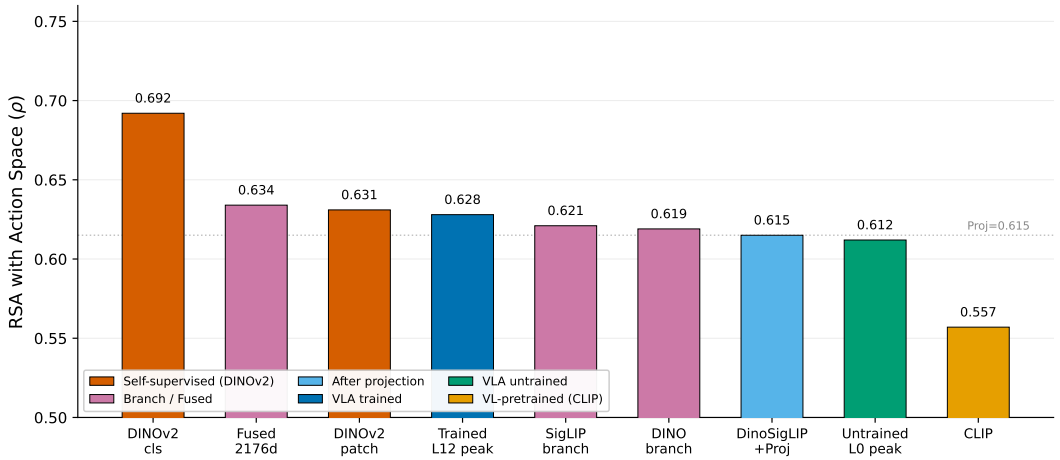


Figure 5: RSA with the action space (ρ) across representation sources. Self-supervised DINOv2 `cls_token` (ρ=0.692) achieves the highest geometric alignment. CLIP (0.557) yields the weakest alignment among pretrained encoders. Error bars: 95% bootstrap CIs.

A.8 Per-Layer RSA Profiles

Figure 6 extends Figure 2 (main text) by adding readout and untrained profiles.

A.9 Per-Layer Action Decoding Across Training Stages

Figure 7 compares per-layer action probe R^2 across three training stages (untrained, Llama-2 base, VLA-trained) using 50 demonstrations per task, averaged over 3 seeds (mean $\pm$ std). All three conditions share L0 representations ($R^2 = 0.645 \pm 0.008$); subsequent differences thus arise from Transformer processing rather than input variation. The trained model peaks at L9 ($R^2 = 0.643 \pm 0.014$), consistent with progressive specialization for action-relevant feature integration. Llama-2 base shows a steady decline (L0: 0.645 $\rightarrow$ L32: 0.551, std 0.003–0.035), consistent with stable but unspecialized processing. The untrained model exhibits high cross-seed variance and deep-layer collapse, with negative R^2 and bimodal behavior across seeds at late layers; this instability indicates that randomly initialized Transformer layers produce destructive interference on action-

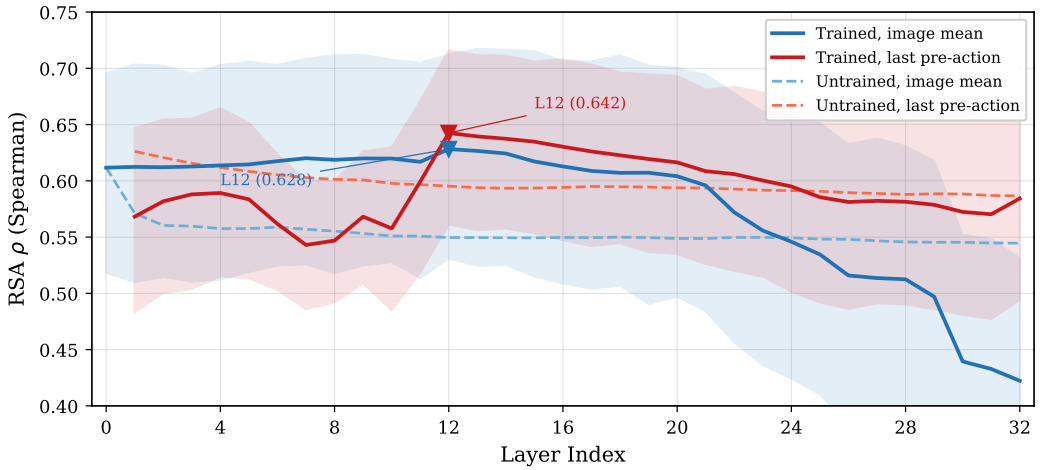


Figure 6: Per-layer RSA (ρ) for trained and untrained models at image_mean and readout (last_preaction) positions. The trained model shows an inverted-U peaking at L12 for both positions; the untrained model shows monotonic decline. After L12, readout and image_mean diverge: readout ρ declines gradually ($0.628 \rightarrow 0.584$) while image_mean ρ drops sharply ($0.628 \rightarrow 0.422$), consistent with task-specific conditioning at the readout position.

relevant structure. Random weights at intermediate layers produce anti-correlated representations ($R^2 < 0$, L10–L11), consistent with untrained networks lacking structured computation. The three-condition gradient (trained $>$ Llama-2 base $>$ untrained) holds from L1 through L15, with a reversal at L32 where trained R^2 drops below the pretrained baseline. The qualitative pattern is consistent across all 3 seeds. Three conditions show substantial agreement in layer-wise R^2 ordering (Kendall's $W = 0.72$, $\chi^2 = 68.99$, $p < 0.001$).

Figure 7: Per-layer action probe R^2 (image_mean) across three training conditions: untrained VLA (random weights), Llama-2 base (pretrained, no VLA fine-tuning), and VLA-trained. All conditions share L0 (0.645 ± 0.008). Trained model peaks at L9; Llama-2 base declines steadily; untrained shows high cross-seed variance and deep-layer collapse. 50 demonstrations per task, 3 seeds (mean $\pm$ std).

A.10 Random-Subset Ablation Details

To evaluate whether the ablation effect in §4.2 is specific to instruction-dominant heads, we compare dominant-head ablation against 30 random subsets of 24/32 L1 heads (each subset drawn without replacement). Dominant-head ablation increases action R^2 by $+0.030$ (within-experiment baseline 0.524), while random subsets yield a mean increase of $+0.071 \pm 0.013$ (Figure 8). The dominant-head effect falls below all 30 random subsets ($z = -3.11$, percentile $= 0\%$); heads with higher instruction-attention fraction carry more organized attention signals, consistent with graded rather than binary specialization.

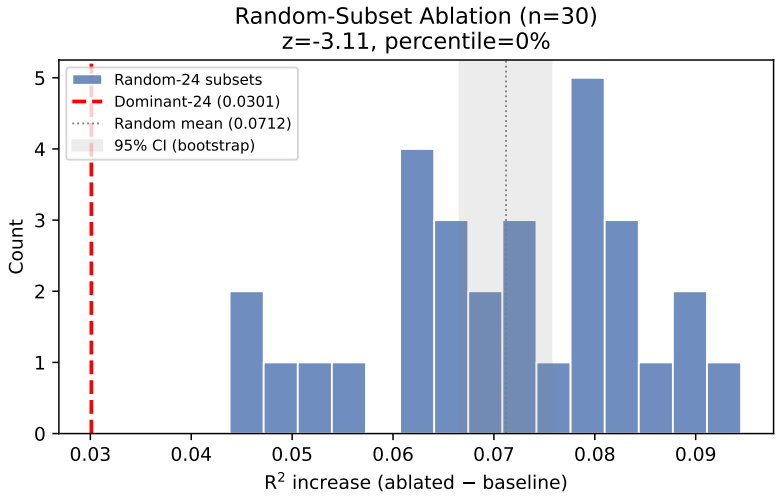


Figure 8: R^2 increase from ablating 24/32 L1 heads. Gray bars: 30 random subsets; red line: dominant-24 heads ($+0.030$). Random subsets yield larger gains (mean $+0.071 \pm 0.013$; $z = -3.11$, percentile $= 0\%$), consistent with graded attention specialization across L1 heads.

A.11 Action Trajectory Analysis Details

Table 7 reports per-dimension mean absolute shift for instruction-dominant (24 heads) and control (8 heads) L1 ablations. Both conditions produce negligible shifts on translational dimensions (dx, dy, $dz < 0.01$) and small rotational shifts (rx, ry, $rz < 0.04$). The gripper dimension absorbs the majority of the effect: instruction-head ablation shifts gripper predictions by 0.291 (control: 0.141), consistent with L1 attention influencing discrete open/close decisions.

Per-task Cohen's d values for instruction-head ablation are 0.43, 0.69, 0.70, 0.70, and 0.73 (mean 0.65), with all tasks significant ($p < 10^{-10}$). Token-level KL divergence between original and ablated action distributions is 5.08 for instruction heads and 1.69 for control heads.

Trajectory-level aggregation. Aggregating per-demo mean actions to the trajectory level reveals that token-level d values are conservative: within-trajectory averaging reduces noise, yielding $\sim 4\times$ larger effect sizes (mean token-to-trajectory ratio 0.245). Instruction-head ablation ($N = 50$: 5 tasks $\times$ 10 demos) yields trajectory $d = 2.943$ (95% CI $[2.469, 3.707]$); bottom-24 non-instruction heads ($N = 30$) yield $d = 2.347$ (CI $[1.854, 3.221]$). All 14 conditions produce trajectory $d > 2.0$ with CI lower bounds > 1.7. Counterfactual trajectory-level d (10 LIBERO-Goal tasks, $N = 100$): wrong 3.798 (CI $[3.194, 4.938]$), empty 3.438 ($[2.921, 4.448]$), shuffled 2.595 ($[2.028, 3.791]$), paraphrased 2.131 ($[1.740, 2.819]$). Token-level d underestimates trajectory-level effects by $\times 3.1$–4.5, consistent with within-trajectory averaging reducing noise. Fine-grained ordering is not reliably distinguished at trajectory level ($N = 100$); token-level comparisons ($N \sim 5{,}000$) provide higher resolution for ordering claims.

Ground-truth action error. Ablation redirects model outputs ($d = 0.65$) but does not systematically increase error relative to expert demonstrations: original MSE $= 0.498$; top-24 instruction-head ablation MSE $= 0.497$ ($\Delta = -0.001$, Wilcoxon $p = 0.52$, n.s.); bottom-24 non-instruction heads MSE $= 0.506$ ($\Delta = +0.008$). Changes concentrate on the gripper dimension; translation and rotation dimensions are negligibly affected. Per-task breakdown shows all $|\Delta\text{MSE}| < 3\%$ of baseline with all $p > 0.27$; the near-zero aggregate Δ reflects uniformly small per-task effects rather than cancellation across tasks. This is consistent with the information-computation dissociation: instruction-dominant heads control instruction-specific direction selection rather than action precision.

Table 4 reports per-head bootstrap CIs (10,000 resamples). Individual control heads produce $\sim 1.9\times$ stronger effects ($d = 0.051$) than instruction heads ($d = 0.027$, $p = 0.0018$), consistent across all 5 tasks. The 24 instruction heads nonetheless produce a larger aggregate perturbation ($d = 0.65$ vs. 0.41), consistent with collective coordination rather than individual strength.

Table 4: Per-head bootstrap analysis of L1 ablation effects. Instruction heads are individually weaker but collectively stronger, consistent with coordinated attention redistribution.

Group	N heads	Collective d	Per-head d [95% CI]
Instruction-dominant	24	0.65	0.027 [0.026, 0.029]
Control	8	0.41	0.051 [0.048, 0.055]
Difference	—	—	0.024, $p = 0.0018$

Instruction-head ablation produces more uniform perturbation across action dimensions (CV $= 1.30$ vs. control CV $= 2.74$, permutation $p = 0.016$), consistent with structured, language-conditioned computation rather than diffuse noise.

Table 5 and Figure 9 report the dose-response gradient for instruction-head ablation at L1. Trajectory divergence increases monotonically from $d = 0.482$ (4 heads) to $d = 0.639$ (32 heads), with steep rise through 12 heads (slope $\Delta d/\Delta 4 \approx 0.026$), a plateau at 16–20 heads ($\Delta d = 0.001$), and slow increase at 20–32.

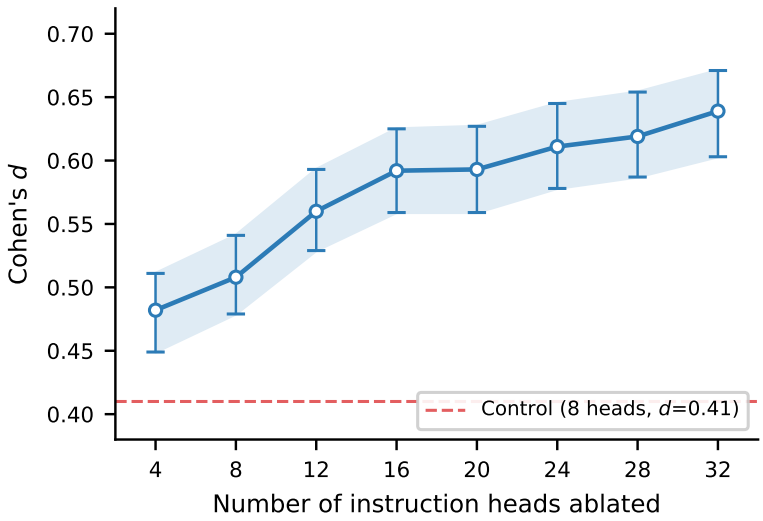


Figure 9: Dose-response gradient: trajectory divergence (Cohen's d) as a function of ablated instruction-dominant heads at L1. Error bars: 95% bootstrap CIs (10,000 resamples). Dashed red: control ablation (8 non-instruction heads, $d = 0.41$). Steep rise through 12 heads, plateau at 16–20, diminishing returns beyond.

Table 5: Dose-response data for L1 instruction-head ablation. Cohen's d vs. original action sequence; 95% bootstrap CIs (10,000 resamples).

Heads ablated	Cohen's d	95% CI
4	0.482	[0.449, 0.511]
8	0.508	[0.479, 0.541]
12	0.560	[0.529, 0.593]
16	0.592	[0.559, 0.625]
20	0.593	[0.559, 0.627]
24	0.611	[0.578, 0.645]
28	0.619	[0.587, 0.654]
32	0.639	[0.603, 0.671]

Table 6: Counterfactual instruction analysis across all 10 LIBERO-Goal tasks. Jaccard: token overlap with original instruction; Cohen's d: mean across 10 tasks. Token novelty ($1 - \text{Jaccard}$) largely predicts divergence ordering ($\rho = 0.8$); the one violation (shuffled $>$ paraphrased) reflects positional sensitivity. L1 ablation included for comparison.

Condition	Jaccard	Cohen's d
Empty (no instruction)	0.000	0.978
Wrong (different task)	0.174	0.924
Shuffled (random order)	1.000	0.763
Paraphrased	0.533	0.707
L1 ablation (24 heads)	—	0.650

A.12 Pretrained LLM Representation Quality Under Visual Input

A potential concern with the three-condition comparison (§3.1) is that the pretrained Llama-2 backbone never encountered projected visual tokens during pretraining, so its representations of such input could be degenerate. We evaluate three diagnostics to assess representation quality. Activation magnitude grows from 16.5 (L0) to 90.2 (L31) for Llama-2 base, compared to $16.5 \rightarrow 51.7$ for the VLA-trained model; the pretrained LLM produces larger but non-exploding activations with no evidence of collapse. Per-layer cosine similarity between L0 and deeper layers decreases progressively from 1.0 to ≈ 0.28 at L25, consistent with gradual differentiation rather than catastrophic out-of-distribution behavior. Top-10 principal component variance explained remains stable across all layers (0.59–0.74), with no low-rank collapse. These diagnostics indicate that the pretrained LLM produces structured, full-rank representations of visual input despite never encountering projected visual tokens during pretraining (Figure 10). These results support the validity of the three-condition comparison.

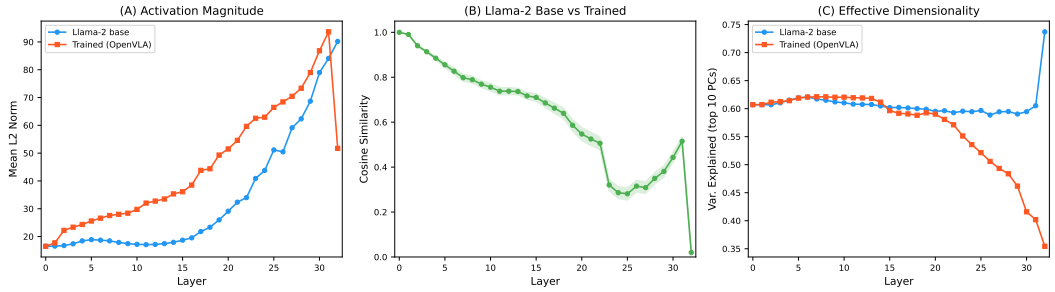


Figure 10: Pretrained LLM representation diagnostics under visual input. Left: activation magnitude across layers (Llama-2 base vs. VLA-trained). Center: cosine similarity between L0 and deeper layers. Right: top-10 PC variance explained. All three metrics indicate structured, non-degenerate representations.

Table 7: Mean absolute action shift per dimension under L1 head ablation. Instruction: 24 dominant heads; Control: 8 remaining heads. Gripper shift dominates both conditions.

Condition	dx	dy	dz	rx	ry	rz	Gripper
Instruction (24)	0.002	0.005	0.008	0.021	0.031	0.034	0.291
Control (8)	0.000	0.001	0.002	0.008	0.014	0.018	0.141

A.13 Probe Architecture Sensitivity

To rule out that our findings depend on MLP probe capacity ($N/\text{params} \approx 0.06$, overparameterized), we replicate the per-layer analysis with a linear probe ($4{,}096 \to 7$, $N/\text{params} \approx 2.2$). The inverted-U pattern reproduces: trained image_mean peaks at L14 ($R^2 = 0.565$), while untrained declines monotonically from L0 ($R^2 = 0.517$). The trained–untrained gap is larger under the linear probe (L14: $\Delta = +0.324$) than the MLP probe (L8: $\Delta = +0.015$); VLA training thus reorganizes representations to make action information more linearly decodable. Untrained deep layers collapse entirely under the linear probe (L25: $R^2 = -0.298$); MLP nonlinearity partially rescues these degraded representations. For last_preaction tokens, the linear probe also corroborates the inverted-U (peak L13, $R^2 = 0.596$), with the same peak layer as the MLP probe and a trained-vs-untrained gap of $\Delta = 0.103$ ($p < 0.001$). These results indicate that our qualitative findings reflect representational structure rather than probe capacity artifacts (Figure 11).

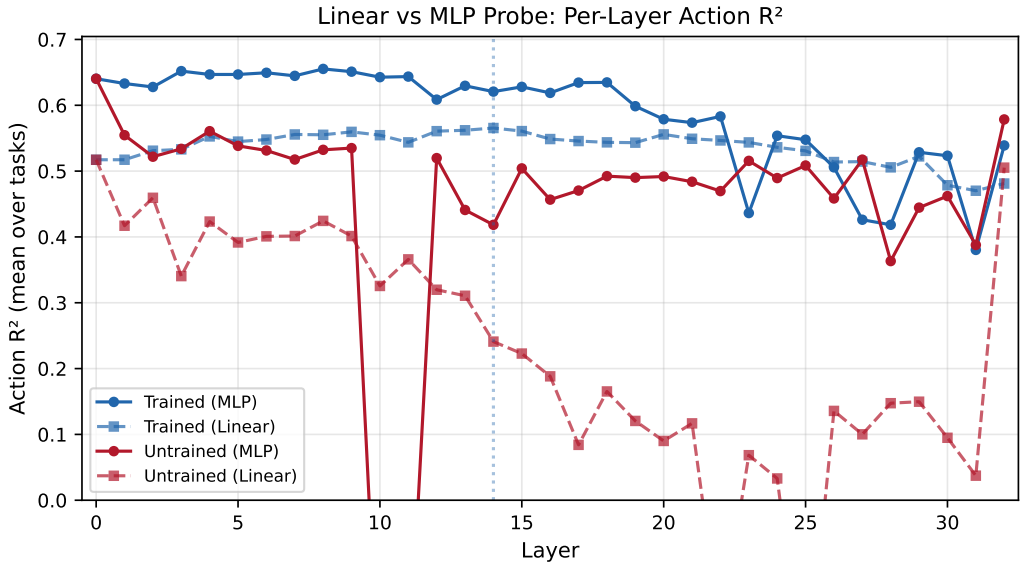


Figure 11: Per-layer action probe R^2 for linear (dashed) and MLP (solid) probes across trained (blue) and untrained (red) conditions. The inverted-U reproduces under both probe architectures; the trained–untrained gap is larger for linear probes.

A.14 Functional Probe Comparison

Figure 12 compares functional probe R^2 across representation sources at each source's best layer.

A.15 Training Effect and Instruction Invariance

Figure 13 shows trained vs. untrained RSA profiles and instruction invariance. Instruction invariance holds across all conditions: maximum $|\Delta| < 0.006$, so action geometry is visually grounded and not modulated by language instructions.

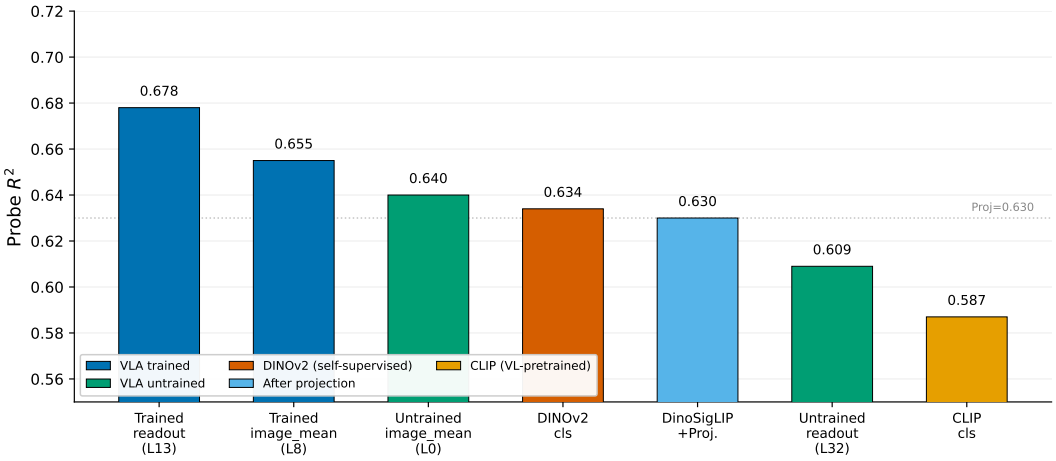


Figure 12: Functional probe R^2 for action prediction (best layer per source). Trained readout (R^2=0.678, L13) and trained image_mean (0.655, L8) exceed the projector baseline (0.630); the inverted-U extends to functional capacity.

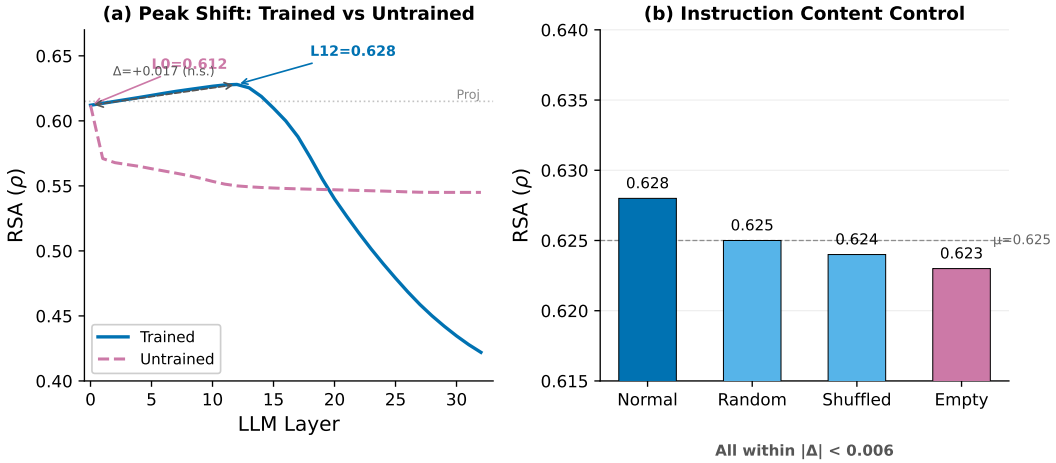


Figure 13: **(a)** Trained vs. untrained image_mean RSA profiles. Training shifts the peak from L0 (0.612) to L12 (0.628), yielding a modest best-vs-best gain ($\Delta = +0.017$, 3%). **(b)** Instruction invariance: all conditions within $|\Delta| < 0.006$; action geometry is visually grounded.

A.16 Additional RSA Values

DINOv2 `patch_mean` achieves RSA $\rho = 0.631$, between the `cls_token` (0.692) and the fused DinoSigLIP (0.634). VLA untrained readout peaks at L1 ($\rho = 0.626$).

A.17 Cross-Benchmark Validation: LIBERO-Spatial

To assess generalization beyond in-distribution tasks, we repeat the core RSA and probe analyses on LIBERO-Spatial (10 out-of-distribution tasks $\times$ 50 demonstrations each, requiring spatial reasoning). Figure 14 shows per-layer RSA and R^2 profiles for both LIBERO-Goal and LIBERO-Spatial side by side.

The inverted-U profile replicates on LIBERO-Spatial: trained image_mean RSA peaks at L14 ($\rho = 0.691$) and R^2 at L13 (0.696), with peaks shifting $\sim$2 layers deeper than LIBERO-Goal (RSA L12, R^2 L8–L10), consistent with increased task difficulty requiring deeper processing. Training produces a significant geometric gain (best-vs-best $\Delta = +0.023$, 95% CI $[0.012, 0.036]$, $p < 0.001$), larger than the LIBERO-Goal effect ($\Delta = +0.017$, CI includes zero).

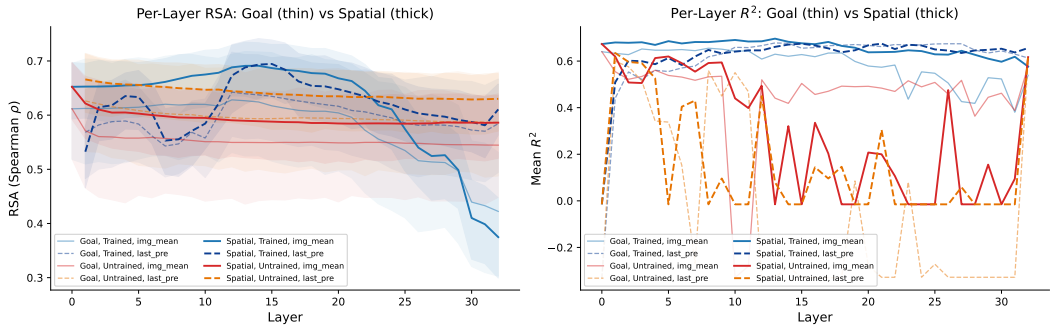


Figure 14: Cross-benchmark validation: per-layer RSA and R^2 profiles on LIBERO-Goal (left) vs. LIBERO-Spatial (right). The inverted-U replicates on both benchmarks. LIBERO-Spatial shows a significant training effect (RSA best-vs-best $\Delta = +0.023$, $p < 0.001$) and deeper peak layers (RSA L14, R^2 L13), but RSA–R^2 peak convergence does not replicate (6-layer gap).

Two notable differences emerge. First, the absolute RSA values are higher on Spatial (0.691 vs. 0.628), possibly reflecting greater action diversity in spatial tasks that produces a more discriminative action RDM. Second, RSA–R^2 peak convergence does not replicate: readout RSA peaks at L15 while R^2 peaks at L21 (6-layer gap), contrasting with the near-alignment on Goal ($|\Delta_{\text{layer}}| = 1$). This suggests that peak convergence is benchmark-specific rather than a general consequence of training.

A.17.1 Cross-Benchmark Counterfactual Replication

Table 8 reports counterfactual instruction analysis on LIBERO-Spatial (10 tasks). The effect ordering is identical to LIBERO-Goal: empty > wrong > shuffled > paraphrased, with the same shuffled–paraphrased violation ($\rho = 0.8$). All effect sizes exceed $d = 0.8$. Spatial instructions differ only in prepositions (e.g., "on the plate" vs. "in the drawer"); the wrong-task $d = 0.851$ demonstrates that instruction-dominant heads encode fine-grained spatial relations, not merely task identity.

Table 8: Counterfactual instruction analysis on LIBERO-Spatial (10 tasks). Effect ordering replicates LIBERO-Goal (Table 6), including the shuffled > paraphrased violation. All $d > 0.8$.

Condition	Jaccard	Cohen's d	CV
Empty (no instruction)	0.000	0.942	0.185
Wrong (different task)	0.174	0.851	0.093
Shuffled (random order)	1.000	0.833	0.166
Paraphrased	0.533	0.806	0.128

A.17.2 Cross-Instruction Probe on LIBERO-Spatial

Table 9 reports cross-instruction 10-way classification accuracy on LIBERO-Spatial. The core pattern from LIBERO-Goal replicates: image_mean remains at chance ($\sim$10%) across all layers and conditions, while the trained readout token saturates to 100% by L4. The untrained readout decays more steeply than on LIBERO-Goal (75.4% at L31 vs. 82%), consistent with greater task difficulty.

Table 9: Cross-instruction 10-way classification accuracy on LIBERO-Spatial. Trained readout achieves 100% from L4; image_mean remains at chance. Untrained readout decays more steeply than on LIBERO-Goal (cf. §4.2).

Token	Condition	L1	L4	L8	L16	L31
last_preaction	Trained	98.9%	100.0%	100.0%	100.0%	100.0%
last_preaction	Untrained	99.5%	98.3%	85.7%	68.3%	75.4%
image_mean	Trained	10.2%	10.0%	9.9%	10.0%	10.1%
image_mean	Untrained	10.1%	10.0%	10.0%	10.0%	10.2%

A.18 Robustness Analyses

A.18.1 Distance Metric Invariance

The inverted-U profile is robust to distance metric choice: cosine action distance yields identical peak layers (L12) and Pearson correlation $r > 0.989$ with the Euclidean profile reported in the main text (Table 10).

Table 10: Peak layer comparison between Euclidean and cosine action distances. Profile r denotes Pearson correlation across all 32 layers.

Condition	Eucl. Peak	Cos. Peak	Profile r
Trained image_mean	L12	L12	0.990
Trained last_preaction	L12	L13	0.990
Untrained image_mean	L0	L0	0.995
Untrained last_preaction	L1	L1	0.992

A.18.2 Per-DOF Decomposition

RSA decomposition by action dimension indicates the inverted-U is not driven by any single degree of freedom: xyz translation (peak L13, $\rho = 0.193$), rotation (peak L12, $\rho = 0.175$), and gripper (peak L13, $\rho = 0.106$) all exhibit non-monotonic profiles. Per-DOF RSA values are lower than the 7-D RSA because single-dimension distances carry less discriminative information than the full action space.

A.18.3 Permutation Test

Permutation testing (10,000 shuffles) indicates all observed RSA values are highly significant: observed ρ (0.42–0.63) exceeds the null distribution maximum (0.08–0.10) by 15–23 standard deviations (all $p < 0.0001$).

A.18.4 Probe Seed Sensitivity

Action probe R^2 is highly stable across random seeds. Over 5 seeds (trained image_mean, full-layer scan), per-layer R^2 standard deviation is uniformly below 0.01 (mean 0.004), and the peak R^2 varies by less than 0.4% (0.655 ± 0.002; Table 11). Peak layer concentrates in the L8–L10 region ($4/5$ seeds); the inverted-U shape is not a seed artifact. Seed 1 peaks at L16, but with R^2 difference <0.002. The inverted-U plateau (L8–L16) is sufficiently flat that argmax is sensitive to minor noise while the profile shape is invariant. A 15-seed replication confirms stability: trained peak remains at L9 ($R^2 = 0.637 \pm 0.004$), with all trained/Llama-2-base layers showing CI width <0.02; untrained exhibits higher variance (std 0.185 at peak), consistent with random initialization instability. Fixed-layer analysis at L9 yields $R^2 = 0.637 \pm 0.008$ (trained) and 0.635 ± 0.011 (Llama-2 base), confirming the inverted-U pattern. The per-seed peak method reports higher absolute values (0.655) because each seed selects its optimal layer independently; both methods agree on the qualitative pattern and the trained $>$ untrained ordering.

Table 11: Probe R^2 peak across 5 random seeds (trained image_mean).

Seed	Peak Layer	Peak R^2
0	L9	0.654
1	L16	0.656
2	L9	0.653
3	L10	0.655
42	L9	0.655
Mean$\pm$std	10.6 ± 2.7	0.655 ± 0.002

A.18.5 Demonstration Subsample Sensitivity

To verify that the inverted-U pattern is not an artifact of sample size, we subsample 25 from 50 demonstrations per task and recompute both RSA and probe profiles (Table 12). The RSA peak layer remains at L12 with a profile correlation of $r = 0.970$ between 50- and 25-demo conditions (high stability). The probe peak shifts from L9 to L11 (within the L8–L16 plateau) with $r = 0.819$; R^2 is more sensitive to training set size than RSA. Both metrics show reduced absolute values (RSA -3.6%, R^2 -7.6%) as expected from halved data, but the non-monotonic profile shape is preserved.

Table 12: Inverted-U stability under demonstration subsampling (trained image_mean).

Metric	50 demos	25 demos	Δ
RSA peak layer	L12	L12	0
RSA peak ρ	0.628	0.606	-3.6%
R^2 peak layer	L9	L11	$+2$
R^2 peak value	0.659	0.609	-7.6%
RSA profile r	—	0.970	—
R^2 profile r	—	0.819	—

A.18.6 Statistical Power

Post-hoc power analysis ($N = 10$ tasks, $\alpha = 0.05$, target power $= 0.80$) indicates minimum detectable effect sizes of $\Delta_{\text{RSA}} = 0.030$ and $\Delta_{R^2} = 0.026$. The observed RSA training effect ($\Delta = 0.017$, Cohen's $d_z = 0.56$) achieves power $= 0.356$, confirming the analysis is underpowered for this effect size; the non-significant RSA result (§4.6) should not be interpreted as evidence of absence. The observed R^2 readout effect ($\Delta = 0.069$, $d_z = 2.68$) achieves power $= 1.000$, confirming adequate sensitivity for the primary functional metric.

A.19 Detailed Training Effects

Table 13 reports bootstrap-based statistical comparisons of training effects on RSA and probe R^2.

Table 13: Training effects on action-relevant structure (LIBERO-Goal; bootstrap, 10,000 resamples). RSA best-vs-best yields a modest geometric gain ($\Delta = +0.017$, 3%; LIBERO-Spatial: $+0.023$, $p<0.001$), while R^2 shows substantially larger functional gains.

Comparison	Δ	95% CI	p
RSA (ρ)			
Best-vs-best image_mean (trained L12 vs. untrained L0)	$+0.017$	incl. 0	n.s.
Same-layer L12 image_mean (trained vs. untrained)	$+0.079$	excl. 0	<0.0001
Instruction effect (max $\lvert\Delta\rvert$)	0.006	—	n.s.
Probe R^2			
Best-vs-best image_mean (trained L8 vs. untrained L0)	$+0.015$	excl. 0	0.006
Best-vs-best readout (trained L13 vs. untrained L32)	$+0.069$	excl. 0	<0.0001

A.20 Training Effect Size Summary

Table 14 consolidates training effects across all metrics. Training effects on image_mean metrics are modest (RSA $\Delta = +0.017$, R^2 $\Delta = +0.015$), but effects on last_preaction metrics are substantial (instruction attention $+18$pp, attention $+46$pp, R^2 $\Delta = +0.069$). This asymmetry indicates that training primarily reshapes how the readout token integrates information rather than altering the visual representations carried by image tokens.

Table 14: Training effect sizes across metrics. Image_mean effects are modest; last_preaction effects are large. Training targets readout-token information integration.

Metric	Trained	Untrained	Δ
RSA peak (image_mean)	0.628 (L12)	0.612 (L0)	+0.017
R^2 peak (image_mean)	0.655 (L8)	0.640 (L0)	+0.015
R^2 peak (last_preaction)	0.678 (L13)	0.609 (L32)	+0.069
Instruction attention L31	100%	82%	+18pp
Instruction attention L1	58%	12%	+46pp
Progress classification	86.3% (L20)	85.3% (L4)	+1pp

A.21 Instruction Classification Probes

A.21.1 Cross-Instruction Classification: Dense Layer Profile

Table 15 shows the dense per-layer profile of cross-instruction classification (10-way) at the readout token for layers L0–L8. The step-function onset between L0 (10%, chance) and L1 ($\geq 99.6\%$) reflects the high discriminability of instruction embeddings from the frozen tokenizer: even incidental attention ($\sim 12\%$) suffices for linear classification at L1. Both pretrained and VLA-trained models sustain 100% through L31; only random initialization shows progressive decay (82% at L31). Deep-layer stability thus derives from structured pretrained weights rather than task-specific fine-tuning.

Table 15: Cross-instruction 10-way classification accuracy at readout token, L0–L8. Values reflect dense classification evaluation; the main text (§4.2) reports sparse single-split results, yielding minor differences (e.g., L8: 98.2% vs. 98.6%).

Layer	Trained	Untrained	Image_mean (trained)
L0	10.0%	10.0%	10.0%
L1	100.0%	100.0%	10.0%
L2	100.0%	99.6%	10.0%
L3	100.0%	99.6%	10.0%
L4	100.0%	99.1%	10.0%
L5	100.0%	98.2%	10.0%
L6	100.0%	97.3%	10.0%
L7	100.0%	96.4%	10.0%
L8	100.0%	98.6%	10.0%

A.21.2 Cross-Instruction LOOCV Validation

Leave-one-out cross-validation over 10 folds (each fold holds out one task's instructions) validates that the cross-instruction probe result is not driven by memorization. Trained last_preaction achieves $100\% \pm 0.00$ across all 10 folds at layers L1–L31 (perfect generalization).

A.21.3 QA Probe: Progress and Subtask Classification

To test whether LLM layers encode task-relevant semantic information beyond instruction identity, we train logistic regression probes (scikit-learn, $C = 1.0$, max_iter $= 1{,}000$) to classify task progress (4-class: each trajectory split into 4 equal temporal segments) and active subtask (binary: first vs. second half) from frozen representations, with an 80/20 random train/test split (Table 16).

Table 16: QA probe accuracy (progress + subtask classification, best layer).

Source	Best Layer	Accuracy
Trained last_preaction	L20	86.3%
Untrained last_preaction	L20	85.3%
DINOv2 cls_token	—	82.7%

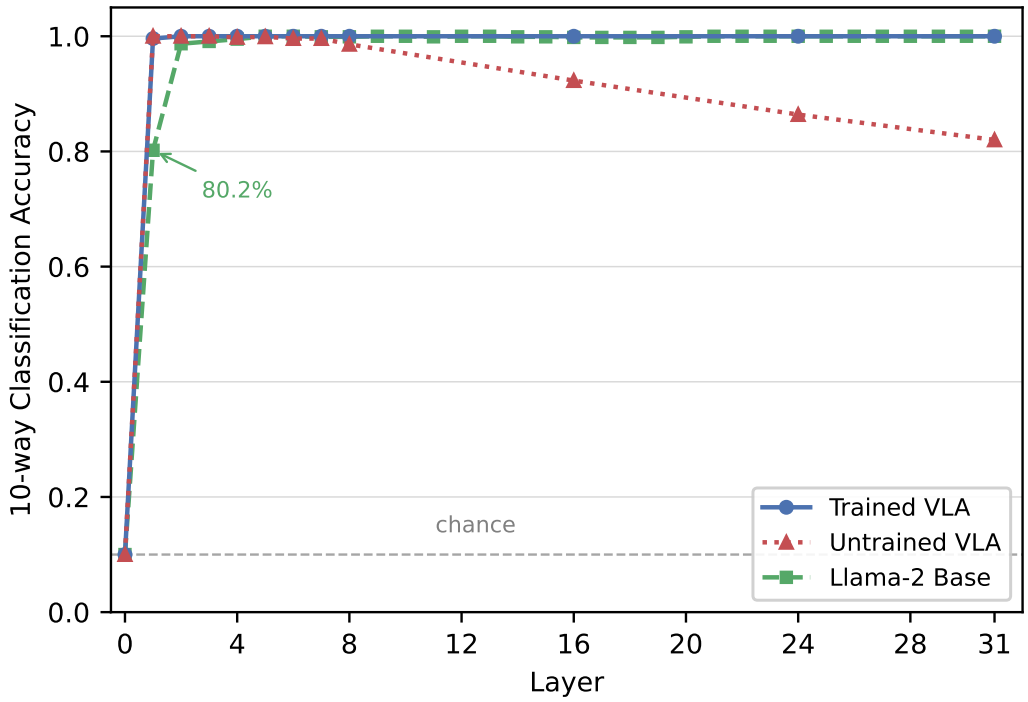


Figure 15: Cross-instruction 10-way classification accuracy at the readout token across all 32 LLM layers. VLA-trained achieves 99.6% at L1, saturating to 100%. Llama-2 base starts at 80.2%, reaching 100% by L5. Untrained starts at 100% but decays to 82% by L31.

A.21.4 Cross-Instruction Probe Profiles

A.21.5 Cross-Architecture Probe and Attention Profile

Table 17 reports cross-instruction probe accuracy and L1 instruction-dominant head counts across six conditions spanning two architectures. The probe-attention dissociation at L1 is notable: TraceVLA training *increases* instruction-dominant heads (28 $\rightarrow$ 32) while *decreasing* probe accuracy (98.1% $\rightarrow$ 83.6%), consistent with representational reorganization for attention efficiency rather than linear separability.

Table 17: Cross-instruction 10-way probe accuracy and L1 instruction-dominant heads across six conditions. Probe accuracy and attention head count are dissociated at L1: TracVLA-Phi3V increases heads (28 $\rightarrow$ 32) while decreasing probe accuracy (98.1% $\rightarrow$ 83.6%), paralleling the information-computation dissociation observed in OpenVLA. Image_mean remains at chance (10%) in all conditions (omitted).

Condition	L1 heads	L1	L2+	L31	Mode
OpenVLA (Llama-2 7B)					
Random init	0/32	100.0%	decays	82.0%	incid.+forget
Llama-2 base	9/32	80.2%	L5: 100%	100.0%	gradual+stable
VLA trained	24/32	99.6%	100%	100.0%	immed.+stable
LIBERO-finetuned	22/32	99.7%	100%	100.0%	immed.+stable
TraceVLA (Phi-3-V 4B)					
Phi-3-Instruct	28/32	98.1%	100%	100.0%	inherited+stable
VLA trained	32/32	83.6%	100%	100.0%	reorg.+stable

A.21.6 L1 Attention Pattern Analysis

Figure 17 shows the per-head L1 attention distribution at the readout token. The three-level gradient (untrained 0/32, Llama-2 base 9/32, VLA-trained 24/32 instruction-dominant heads) shows that

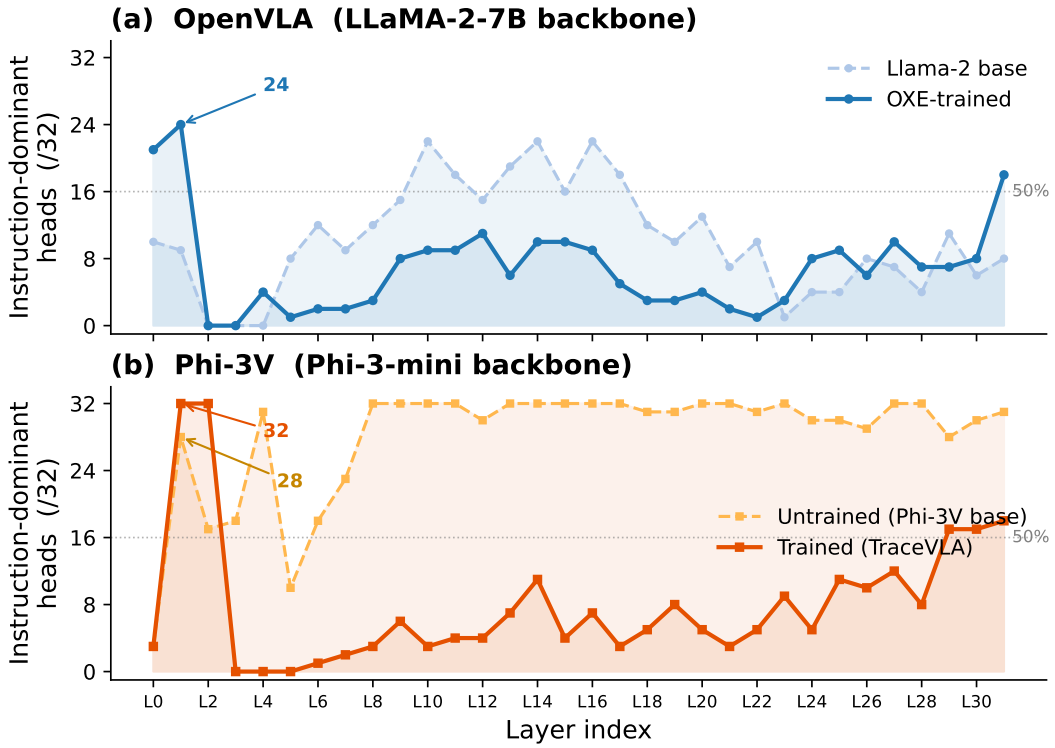


Figure 16: Instruction-dominant heads across layers for two architectures (OpenVLA / TraceVLA), each comparing pre-VLA and VLA-trained conditions. Despite opposite starting points—OpenVLA (Llama-2) begins with sparse attention while TraceVLA (Phi-3-V) begins with distributed attention—VLA training drives both to entry-exit concentration (L1 saturation, mid-layer suppression, L29–L31 re-engagement).

language pretraining seeds partial instruction attention and VLA fine-tuning shifts it further toward instruction dominance.

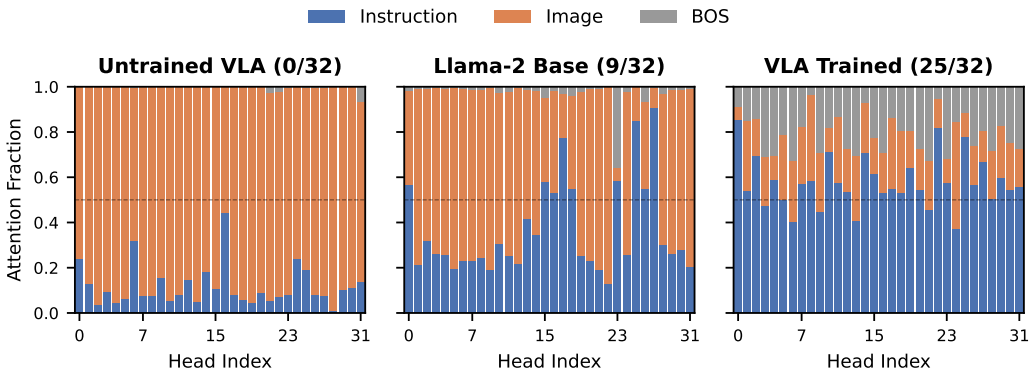


Figure 17: L1 attention from the readout token across three conditions, per head (sorted by instruction attention fraction). VLA-trained: 24/32 instruction-dominant heads (57.7% text attention). Llama-2 base: 9/32 heads (36.5%). Untrained: 0/32 heads (11.7%). See Table 18 for quantitative comparison across layers.

A.21.7 Multi-Layer Attention Patterns

Table 18 shows a 4-layer subset of the full 32-layer attention profile. The full profile (Figure 18) reveals three qualitatively distinct patterns: the untrained model maintains 0 instruction-dominant heads at all 32 layers; Llama-2 base shows a mid-depth peak (L9–L16: 15–22 heads), consistent

with typical autoregressive LM processing depth; VLA fine-tuning produces an entry-exit pattern (L0–L1: 21–24 heads, L31: 18 heads, intermediate layers 0–11).

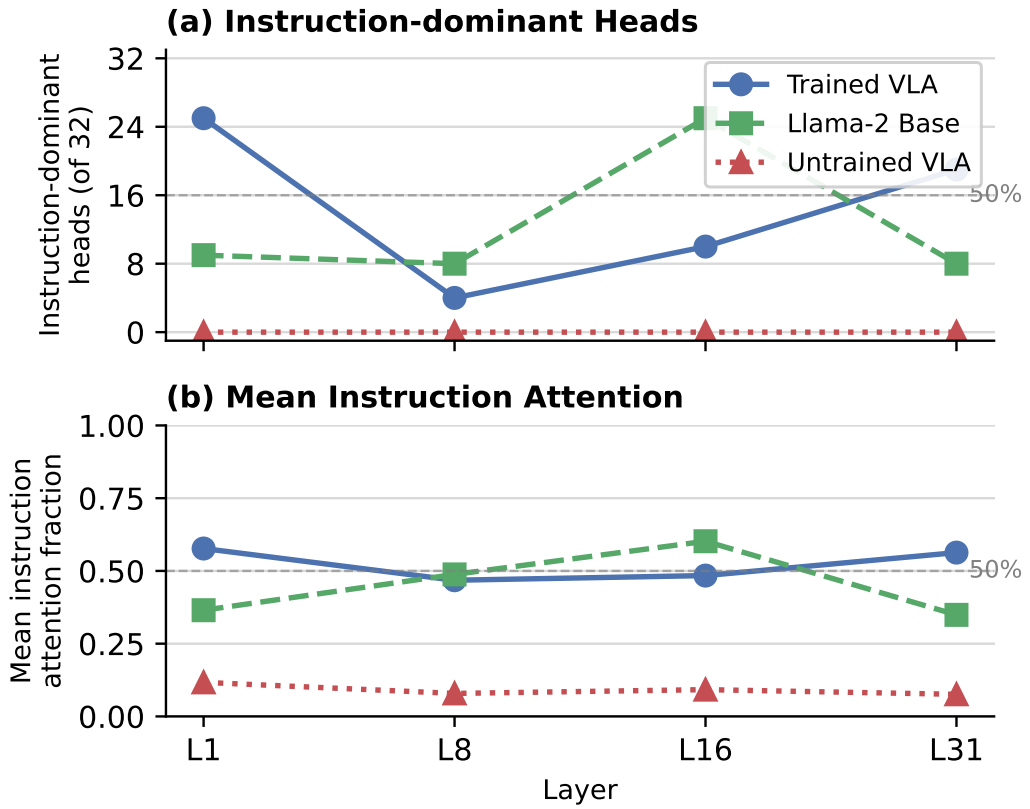


Figure 18: Instruction-dominant heads and mean text attention fraction across layers for three conditions. VLA-trained shows elevated values at L1 and L31; Llama-2 base peaks at L16; untrained remains near zero throughout.

Table 18: Instruction-dominant heads and text attention fraction at four sampled layers (subset of the 32-layer profile in Figure 18). VLA-trained shows entry-exit concentration (L1, L31); Llama-2 base peaks at mid-depth (L16); untrained shows no instruction attention. Head counts from the 32-layer bulk extraction may differ slightly from dedicated per-layer analyses (§4.2) due to threshold boundary effects.

	VLA Trained		Llama-2 base		Untrained	
Layer	Heads	Text%	Heads	Text%	Heads	Text%
L1	**24/32**	57.7	9/32	36.5	0/32	11.7
L8	4/32	46.8	8/32	48.8	0/32	7.9
L16	10/32	48.4	**25/32**	60.2	0/32	9.2
L31	**18/32**	56.3	8/32	34.9	0/32	7.6

A.21.8 Threshold Sensitivity Analysis

The main text defines instruction-dominant heads as those allocating $> 50\%$ attention to instruction tokens. Table 19 shows that the three-condition gradient (untrained $<$ Llama-2 base $<$ VLA-trained) holds across all thresholds from 30% to 70%; the 50% threshold is not a boundary-sensitive choice.

A.22 L1 Attention Decomposition by Head Group

Table 20 decomposes L1 attention mass at the readout token into three token categories for instruction-dominant and control head groups.

Table 19: Number of instruction-dominant heads at L1 under varying attention thresholds. The three-condition ordering is stable across all thresholds.

Threshold	VLA Trained	Llama-2 base	Untrained
30%	32	13	0
40%	31	10	0
50%	24	9	0
60%	9	4	0
70%	5	3	0

Table 20: Attention mass allocation at Layer 1, averaged across 5 samples. Instruction-dominant heads (defined as $> 50\%$ attention to instruction tokens) allocate 17.5 percentage points more attention to instruction tokens than control heads, with corresponding reductions in vision (-8.3pp) and BOS/sink (-9.2pp) attention.

Head Group	N	Instruction	Vision	BOS/Other
Instruction-dominant	24	62.4%	18.8%	18.8%
Control	8	44.9%	27.1%	28.0%
Δ	—	$+17.5$pp	-8.3pp	-9.2pp

Head-level instruction attention entropy reaches 99.5% of the uniform maximum; attention is broadly distributed rather than concentrated in a few heads.

Per-token attention normalization. The 57.7% aggregate text attention in instruction-dominant heads partly reflects token-count imbalance (22 instruction vs. 256 image tokens). Per-token normalization reveals a $38.2\times$ ratio: each instruction token receives $38.2\times$ the attention of each image token in trained instruction-dominant heads, compared to only $1.5\times$ in untrained heads (a $25.5\times$ increase from training). Instruction-dominant heads also show lower attention entropy (0.465) than non-dominant heads (0.569), indicating more concentrated attention distributions.

A.23 Per-Dimension Action MAE Under Head Ablation

Table 21 extends Table 7 with full per-dimension control-head data and instruction-to-control MAE ratios.

Table 21: Mean absolute error per action dimension under L1 head ablation. Translation dimensions (dx/dy/dz) show 3–5$\times$ larger instruction-to-control ratio than rotation dimensions (1.9–2.6$\times$); instruction-dominant heads have disproportionate influence on positional control. Gripper (2.07$\times$) falls between the two groups.

Dimension	Instr. 24h MAE	Control 8h MAE	Ratio
dx	0.00189	0.00037	5.09×
dy	0.00531	0.00125	4.26×
dz	0.00762	0.00243	3.14×
$roll$	0.02059	0.00785	2.62×
$pitch$	0.03071	0.01387	2.21×
yaw	0.03388	0.01783	1.90×
$gripper$	0.29072	0.14072	2.07×

A.24 Per-Task R^2 Distribution Under L1 Ablation

Table 22 reports per-task action probe R^2 at L1 under baseline and 24-head ablation conditions (MLP probe, 5 demonstrations per task).

A.25 Task Complexity and Attention Dependence

We classify tasks as *instruction-discriminative* when multiple tasks share the same manipulated object(s), requiring instruction content to disambiguate the target action (e.g., "put bowl on plate"

Table 22: Per-task action probe R^2 at L1 (MLP probe, 5 demos/task). All 10 tasks show positive ablation effect ($\Delta > 0$): instruction-head ablation uniformly increases action decodability at the entry layer. Mean $\Delta R^2 = +0.107$; the 50-demo setting (§4.2) yields a consistent $+0.059$.

Task	Baseline R^2	Ablated R^2	ΔR^2
open middle drawer	0.197	0.221	+0.024
open top drawer + bowl	0.265	0.340	+0.075
push plate to stove	0.190	0.197	+0.007
put bowl on plate	0.375	0.474	+0.099
put bowl on stove	0.235	0.395	+0.160
put bowl on cabinet	0.256	0.457	+0.201
put cream cheese in bowl	0.186	0.345	+0.159
put wine on rack	0.189	0.424	+0.235
put wine on cabinet	0.279	0.353	+0.074
turn on stove	0.385	0.421	+0.036
Mean $\pm$ std	0.256 ± 0.070	0.363 ± 0.088	+0.107

vs. "put bowl on stove"); remaining tasks involve unique object-action pairings recoverable from visual context alone. Table 23 reports per-task $|\Delta R^2|$ under both groupings.

Table 23: Task-level L1 ablation effects ($|\Delta R^2|$) grouped by instruction dependence. *Primary* (conservative): cream cheese in bowl classified as visual-sufficient ($2.5\times$, $p = 0.033$). *Sensitivity* (shared-object criterion): cream cheese reclassified as instruction-discriminative ($6.5\times$, $p = 0.008$).

Task	$\lvert\Delta R^2\rvert$	Group
put wine on rack	0.235	Instr-disc
put bowl on cabinet	0.201	Instr-disc
put bowl on stove	0.160	Instr-disc
put bowl on plate	0.099	Instr-disc
open top drawer + bowl	0.075	Instr-disc
put wine on cabinet	0.074	Instr-disc
put cream cheese in bowl†	0.159	Vis-suff
turn on stove	0.036	Vis-suff
open middle drawer	0.024	Vis-suff
push plate to stove	0.007	Vis-suff
Primary: Instr-disc mean 0.141; Vis-suff mean 0.057		
Mann-Whitney $U = 21.0$, $p = 0.033$, ratio $= 2.49\times$		
Sensitivity†: cream cheese as Instr-disc: $p = 0.008$, $6.5\times$		

Leave-one-out sensitivity. All leave-one-out p-values remain ≤ 0.083 (marginally significant), with no single task removal reversing the effect direction. Reclassifying the boundary case (cream cheese in bowl) to the instruction-discriminative group strengthens the result ($p = 0.008$).

Bootstrap 95% CI on the complex/simple ratio: $[1.03, 11.12]$ (10,000 resamples). The wide interval reflects low statistical power from $N = 10$ tasks split into groups of 6 and 4; the point estimate ($2.5\times$) persists under leave-one-out analysis (all $p \leq 0.083$, marginally significant; §A.25).

A.26 Specificity Control: Matched-Size Head Ablation

To test whether ablation effects are specific to instruction-dominant heads, we ablate four matched-size subsets of 24 heads at L1: instruction-dominant (highest instruction attention fraction), image-dominant (highest image attention fraction), random (5 seeds), and bottom-24 (lowest instruction attention fraction; overlaps 16/24 heads with the instruction-dominant group).

All conditions yield comparable ground-truth MSE (0.487–0.504 vs. baseline 0.497), consistent with direction-selective rather than precision-degrading effects.

Table 24: Matched-size ablation controls: trajectory disruption (d) from ablating 24-head subsets at L1, ranked by instruction attention fraction. Instruction-dominant heads produce the largest effect, but the modest gap (8% over random) indicates distributed, attention-graded redistribution rather than a dedicated subset.

Condition (24 heads)	Cohen's d	95% CI
Instruction-dominant	0.650	(reference)
Random (5 seeds)	0.601 ± 0.021	$[0.568, 0.622]$
Image-dominant	0.580	$[0.547, 0.613]$
Bottom-24	0.565	—

A.27 Paired Effect Sizes

Main-text Cohen's d values use the standard (unpaired) formula for token-level comparisons where observations are independent tokens. For within-subject effects where the same tasks or demonstrations are measured under two conditions, we additionally report paired $d_z = \bar{\Delta}/s_\Delta$, which accounts for within-subject correlation and yields larger effect sizes when correlation is high (Table 25).

Table 25: Paired effect sizes (d_z) for within-subject comparisons. Standard (unpaired) d reported where available. Paired d_z exceeds standard d when within-subject correlation is high.

Comparison	Standard d	Paired d_z	N
L1 ablation: 24 dominant vs. baseline	0.65	1.17	10 tasks
Random-24 ablation vs. baseline	—	1.84 ± 0.33	30 subsets
Dose-response: 32 vs. 4 heads (MSE)	—	1.79	10 tasks
RSA: trained vs. untrained (best-vs-best)	—	0.55	10 tasks
Cross-arch: OpenVLA vs. TraceVLA peak	—	0.48	15 seeds

A.28 Cross-Architecture Action Decodability: 2×2 Design

Table 27 reports per-layer action prediction R^2 across two architectures (OpenVLA, TraceVLA) $\times$ two training conditions (VLA-trained, untrained base LLM), providing a 2×2 factorial view of how VLA training reshapes per-layer action decodability.

Convergent reorganization from divergent starting points. Untrained profiles differ qualitatively: OpenVLA (Llama-2) peaks at L0 ($R^2 = 0.629$) then collapses to near-zero at mid-layers (L20: 0.086, std $= 0.311$), while Phi-3-V peaks at L6 (0.578 ± 0.018) with gradual monotonic decline (late-layer std $= 0.068$). After VLA training, both architectures reach nearly identical mid-layer peaks—OpenVLA at L9 ($R^2 = 0.637 \pm 0.008$), TraceVLA at L11 (0.634 ± 0.009)—with $\Delta = 0.003$ and overlapping CIs. Deep-layer variance drops sharply with training: OpenVLA std $0.277 \to 0.040$; Phi-3-V std $0.068 \to 0.011$. Peak R^2 values are comparable before and after training (untrained peaks reach 91–99% of trained), but peak locations shift deeper (L0→L9; L6→L11) and profile shapes transform from architecture-specific patterns to a shared mid-layer plateau.

Summary. Table 26 condenses the key contrasts. The L0 values are identical between trained and untrained within each architecture (shared frozen vision encoder + projector), confirming that training effects arise from LLM weight changes. The largest training gains appear at mid-to-deep layers (L15–L25 average: OpenVLA $+0.313$, Phi-3-V $+0.103$), precisely where untrained profiles diverge most.

Equivalence testing. To formally test whether the cross-architecture peak R^2 values are statistically equivalent (rather than merely failing to differ), we apply the Two One-Sided Tests (TOST) procedure [46] with equivalence bound $\varepsilon = 0.05$. The observed difference is $\Delta = 0.003$ (OpenVLA L9: 0.637 ± 0.008; TraceVLA L11: 0.634 ± 0.009). Both one-sided tests reject at $p < 0.0001$; the 90% confidence interval for the difference is $[-0.002, 0.009]$, falling entirely within $[-0.05, 0.05]$. A conventional two-sample t-test yields $p = 0.318$ (not significant), illustrating the distinction

Table 26: R^2 profile summary across architectures and training conditions.

	OpenVLA (Llama-2-7B)		TraceVLA (Phi-3-mini)	
	Untrained	Trained	Untrained	Trained
Peak Layer	L0	L9	L6	L11
Peak R^2	0.629	0.637	0.578	0.634
Profile Shape	Collapse	Flat plateau	Ramp $\rightarrow$ decline	Ramp $\rightarrow$ plateau
Late-layer std	0.277	0.040	0.068	0.011
Δ_{train} (peak)	$+0.008$ (L0$\rightarrow$L9)		$+0.056$ (L6$\rightarrow$L11)	
Δ_{train} (L15–L25 avg)	$+0.313$		$+0.103$	

between absence of evidence for a difference and evidence of equivalence: the TOST procedure provides the latter.

A.29 Global Multiple Comparison Correction

The paper reports 63 statistical tests spanning permutation tests, t-tests, bootstrap comparisons, TOST equivalence, and Spearman correlations. We apply a Bonferroni correction ($\alpha_{\text{adj}} = 0.05/63 = 7.94 \times 10^{-4}$) and the less conservative Holm–Bonferroni step-down procedure; both yield identical results. Of the 45 originally significant tests, 40 survive correction (88.9%).

All eight core tests supporting the paper's main claims remain significant: RSA permutation at L12 ($p < 0.0001$), trained vs. untrained image geometry at L12 ($p = 3.9 \times 10^{-5}$), counterfactual wrong and empty instruction ($p < 10^{-100}$), TOST cross-architecture equivalence ($p < 0.0001$), dose-response top-24 ablation ($p < 10^{-100}$), trained vs. untrained probe R^2 ($p < 0.0001$), and bottom-24 specificity ($p < 10^{-90}$).

Five auxiliary tests lose significance under correction: (i) RSA trained vs. untrained preaction at L12 ($p = 0.005 \rightarrow p_{\text{adj}} = 0.284$), (ii) DINOv2 vs. SigLIP RSA ($p = 0.032 \rightarrow p_{\text{adj}} = 1.0$), (iii) fused backbone vs. SigLIP RSA ($p = 0.011 \rightarrow p_{\text{adj}} = 0.673$), (iv) CV exact paired permutation ($p = 0.016 \rightarrow p_{\text{adj}} = 0.984$), and (v) R^2 projector vs. VLA Spearman correlation ($p = 0.003 \rightarrow p_{\text{adj}} = 0.183$). These concern auxiliary comparisons (encoder ranking, coefficient-of-variation ordering) rather than the core redistribution, probing, or counterfactual findings; no main-text conclusion requires revision.

A.30 Mechanistic Interpretation

Matched-size ablation (d: instruction 0.65, random 0.60, image 0.58; Table 24) and dose-response (steep through 12 heads, plateau beyond 20) indicate distributed attention redistribution. Counterfactual ordering largely follows token novelty ($\rho = 0.8$; §4.2), with one violation: shuffled tokens exceed paraphrased ($d = 0.763$ vs. 0.707) despite identical token sets, indicating positional sensitivity. That ablation raises R^2 yet alters trajectories ($d = 0.65$) without increasing ground-truth error ($\Delta\text{MSE} \approx 0$; §A.11) reflects an information-computation dissociation [47]: instruction-dominant heads control direction rather than precision. Random ablation yields larger R^2 gains ($+0.071$ vs. $+0.030$; $z = -3.11$, 0th percentile), consistent with distinct R^2 signatures.

A.31 Functional Stages and Predictions

Per-layer profiles suggest three stages: redistribution (L0–L1; probe 99.6%, 24/32 instruction-dominant heads), computation (L8–L13; R^2 peak 0.655–0.678), and decay (L14–L31, $\sim$56% of parameters; declining R^2 and RSA suggest pruning potential). Task-specific fine-tuning eliminates L31 re-engagement (18 $\rightarrow$ 0 heads; §4.2), consistent with exit-layer attention serving multi-task flexibility (N=2). TraceVLA [8] (Phi-3-V) confirms consistent entry-exit attention concentration from an opposite starting point (§4.2). The 2×2 trained-vs-untrained comparison reveals a consistent reorganization pattern: untrained OpenVLA collapses from L0 ($R^2 = 0.629$) to near-zero at mid-layers (L20: 0.086, std = 0.277), while untrained Phi-3-V peaks at L6 (0.578) with gradual decline (std

Table 27: Per-layer action prediction R^2: training effect across architectures (mean $\pm$ std over 15 seeds). Peak values per condition in **bold**.

Layer	OpenVLA (Llama-2-7B)		TraceVLA (Phi-3-mini)	
	Untrained	Trained	Untrained	Trained
L0	$\mathbf{0.629 \pm 0.016}$	0.629 ± 0.016	-0.330 ± 0.003	-0.330 ± 0.002
L1	0.539 ± 0.023	0.625 ± 0.017	0.429 ± 0.034	0.447 ± 0.013
L2	0.517 ± 0.047	0.622 ± 0.010	0.458 ± 0.023	0.522 ± 0.014
L3	0.448 ± 0.216	0.624 ± 0.014	0.498 ± 0.036	0.579 ± 0.011
L4	0.496 ± 0.057	0.628 ± 0.014	0.572 ± 0.021	0.592 ± 0.009
L5	0.475 ± 0.101	0.627 ± 0.019	0.573 ± 0.020	0.602 ± 0.010
L6	0.422 ± 0.119	0.631 ± 0.012	$\mathbf{0.578 \pm 0.018}$	0.609 ± 0.012
L7	0.459 ± 0.093	0.632 ± 0.012	0.569 ± 0.023	0.616 ± 0.014
L8	0.347 ± 0.283	0.635 ± 0.013	0.560 ± 0.025	0.615 ± 0.010
L9	0.445 ± 0.127	$\mathbf{0.637 \pm 0.008}$	0.564 ± 0.021	0.617 ± 0.013
L10	0.353 ± 0.224	0.624 ± 0.014	0.551 ± 0.027	0.625 ± 0.007
L11	0.356 ± 0.122	0.623 ± 0.016	0.561 ± 0.025	$\mathbf{0.634 \pm 0.009}$
L12	0.409 ± 0.203	0.626 ± 0.011	0.557 ± 0.026	0.632 ± 0.005
L13	0.341 ± 0.227	0.624 ± 0.013	0.547 ± 0.028	0.633 ± 0.008
L14	0.374 ± 0.211	0.623 ± 0.015	0.550 ± 0.023	0.631 ± 0.009
L15	0.297 ± 0.225	0.619 ± 0.012	0.550 ± 0.022	0.633 ± 0.010
L16	0.323 ± 0.226	0.619 ± 0.011	0.534 ± 0.018	0.631 ± 0.010
L17	0.395 ± 0.109	0.609 ± 0.023	0.519 ± 0.024	0.633 ± 0.007
L18	0.214 ± 0.307	0.613 ± 0.017	0.502 ± 0.047	0.629 ± 0.012
L19	0.241 ± 0.258	0.613 ± 0.013	0.499 ± 0.058	0.627 ± 0.008
L20	0.086 ± 0.311	0.611 ± 0.014	0.482 ± 0.048	0.628 ± 0.011
L21	0.330 ± 0.216	0.589 ± 0.023	0.476 ± 0.036	0.624 ± 0.007
L22	0.207 ± 0.283	0.586 ± 0.013	0.408 ± 0.140	0.616 ± 0.008
L23	0.264 ± 0.227	0.578 ± 0.028	0.455 ± 0.074	0.613 ± 0.007
L24	0.173 ± 0.275	0.577 ± 0.013	0.446 ± 0.062	0.610 ± 0.008
L25	0.130 ± 0.305	0.553 ± 0.053	0.479 ± 0.039	0.609 ± 0.010
L26	0.251 ± 0.254	0.535 ± 0.046	0.411 ± 0.098	0.611 ± 0.008
L27	0.099 ± 0.278	0.552 ± 0.027	0.366 ± 0.197	0.604 ± 0.011
L28	0.099 ± 0.307	0.544 ± 0.034	0.358 ± 0.218	0.596 ± 0.015
L29	0.088 ± 0.275	0.530 ± 0.034	0.356 ± 0.271	0.585 ± 0.010
L30	0.220 ± 0.234	0.511 ± 0.029	0.424 ± 0.104	0.574 ± 0.008
L31	0.069 ± 0.284	0.488 ± 0.054	0.332 ± 0.262	0.557 ± 0.016
L32	0.584 ± 0.011	0.535 ± 0.015	0.410 ± 0.185	0.550 ± 0.007

$= 0.068$)—qualitatively opposite untrained landscapes. Yet training drives both to nearly identical mid-layer peaks (0.637 vs. 0.634, $\Delta = 0.003$) with stabilized deep-layer variance (OpenVLA std $0.277 \rightarrow 0.040$; Phi-3-V $0.068 \rightarrow 0.011$). This consistent outcome from divergent starting points suggests that mid-layer action specialization is a consistent convergence pattern of VLA training, not an architectural artifact. Exit-layer R^2 differs (0.488 vs. 0.557), possibly reflecting architectural differences in late-layer processing. Remaining predictions: cross-attention VLAs (π_0 [12], Octo [44]) should lack L1 instruction-dominant heads; larger VLAs should yield flatter per-head contributions. For deployments, L1 instruction probes can flag instruction-discriminative tasks and L1 attention entropy can monitor instruction under-attendance at runtime.

A.32 Extended Discussion

Observational three-tier gradient. The cross-architecture matrix suggests a three-tier gradient of representational adaptation. Tier 1 (VLM bases, no action training): inherited L1 attention (TracVLA 32/32, NORA 13–14/16), negligible exit collapse ($\Delta R^2 \leq 0.048$), and minimal counterfactual sensitivity (cosine distance ≤ 0.185). Tier 2 (general-purpose VLA, OXE-pretrained): strong L1 routing (24–26/32 heads), moderate counterfactual sensitivity ($d \approx 1.0$), and no systematic exit collapse across suites. Tier 3 (task-specific VLAs, fine-tuned on LIBERO): exit collapse at every VLM-to-VLA cell (6/6, $p = 0.016$) and substantial counterfactual sensitivity ($d \geq 0.45$). This gradient is suggestive but limited: Tier 2 contains a single architecture (OpenVLA), and the Tier 2/Tier 3 boundary confounds training stage with architecture. Testing additional general-purpose VLAs (e.g., Octo [44], π_0 [12]) at Tier 2 is necessary before the three-tier structure can be treated as a robust empirical pattern.

Behavioral relevance. Correlating trained ΔR^2 with published LIBERO success rates across all nine available architecture-suite cells yields a positive but non-significant trend (Spearman $\rho = 0.583$, $p = 0.10$, $n = 9$), marginally above the conventional significance threshold. Within-suite comparisons, which control for task difficulty, are uniformly positive: on Goal ($n = 3$ architectures), ΔR^2 and success rate are perfectly rank-correlated ($\rho = 1.0$), and every remaining suite ($n = 2$ each) shows the same direction. This directional evidence is consistent with stronger mid-network action encoding (higher peak R^2) followed by more aggressive exit-layer compression being associated with better task performance. Several caveats temper this interpretation: architecture capacity differences (model size, backbone quality) are not controlled, suite difficulty co-varies with both ΔR^2 and success rate, multiple suites from the same architecture are not independent observations, and the sample size limits statistical power. This correlational evidence does not establish a causal link between exit collapse and task success. Closed-loop behavioral validation, including intervention experiments that directly manipulate exit-layer representations, remains a priority for future work.

Metric constraints. Counterfactual sensitivity is measured via Cohen's d on action MSE for models that produce actions and via cosine distance on hidden representations for untrained VLMs (§4.4). These metrics differ in scale and distributional properties; ratios between them are not meaningful. The OpenVLA Long-Horizon inversion (trained $d = 0.56$, untrained $d = 1.05$) illustrates how same-metric comparisons can reverse the expected direction, cautioning against overinterpretation of any single cell. A standardized metric framework for cross-condition counterfactual comparison remains an open problem.

Extended limitations. Each architecture is analyzed at a single trained checkpoint, precluding characterization of how exit collapse or counterfactual sensitivity emerge during training. All analyses are open-loop, consistent with prior representational studies [5, 6]; closed-loop validation would provide complementary evidence about the behavioral relevance of the identified routing structures. The three-tier gradient has $n = 1$ at Tier 2, limiting its inferential power; populating this tier with additional general-purpose VLAs is a priority for future work.